AMERICAN COLLEGE OF SPORTS MEDICINE

Published ahead of Print

Assessing the Validity of Wearable Inertial Sensors in Evaluating Joint Kinetics and Hamstring Musculotendon Mechanics at Various Running Speeds

Yi-Chung Lin[1,2], Kara Price[1,2], Declan Carmichael[1,2,6], Nirav Maniar[1,2], Jack Hickey[2,3], Ryan Timmins[1,2], Bryan Heiderscheit[4], Silvia Blemker[5,6], and David Opar[1,2]

[1]School of Behavioural and Health Sciences, Australian Catholic University, Melbourne, AUSTRALIA; [2]Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, AUSTRALIA; [3]Department of Sport Science and Nutrition, Maynooth University, Co. Kildare, IRELAND; [4]Department of Orthopedics and Rehabilitation, University of Wisconsin-Madison, Madison, WI; [5]Department of Biomedical Engineering, University of Virginia, Charlottesville, VA; [6]Springbok Analytics, Charlottesville, VA

Accepted for Publication: 9 June 2025

ACCEPTED

Medicine & Science in Sports & Exercise® **Published ahead of Print** contains articles in unedited manuscript form that have been peer reviewed and accepted for publication. This manuscript will undergo copyediting, page composition, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered that could affect the content.

Copyright © 2025 American College of Sports Medicine

Medicine & Science in Sports & Exercise, Publish Ahead of Print
DOI: 10.1249/MSS.0000000000003786

Assessing the Validity of Wearable Inertial Sensors in Evaluating Joint Kinetics and Hamstring Musculotendon Mechanics at Various Running Speeds

Yi-Chung Lin[1,2], Kara Price[1,2], Declan Carmichael[1,2,6], Nirav Maniar[1,2], Jack Hickey[2,3], Ryan Timmins[1,2], Bryan Heiderscheit[4], Silvia Blemker[5,6], and David Opar[1,2]

[1]School of Behavioural and Health Sciences, Australian Catholic University, Melbourne, AUSTRALIA; [2]Sports Performance, Recovery, Injury and New Technologies (SPRINT) Research Centre, Australian Catholic University, Melbourne, AUSTRALIA; [3]Department of Sport Science and Nutrition, Maynooth University, Co. Kildare, IRELAND; [4]Department of Orthopedics and Rehabilitation, University of Wisconsin-Madison, Madison, WI; [5]Department of Biomedical Engineering, University of Virginia, Charlottesville, VA; [6]Springbok Analytics, Charlottesville, VA

Address for correspondence: Yi-Chung Lin, 17 Young Street, Fitzroy, VIC, Australia, 3065; E-mail: yi-chung.lin@acu.edu.au; Phone: +61 3 9953 3772

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Conflict of Interest and Funding Source: The study was funded by the National Football League and the Study Melbourne Research Partnerships Program (Grant Number: 44012).

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

ABSTRACT

Purpose: Integrating musculoskeletal (MSK) modelling with inertial measurement units (IMUs) offers a promising approach for analysing joint and muscle function during locomotion. This study examined the validity of combining IMUs, MSK modelling, and inverse dynamics to estimate lower-limb joint moments and hamstring musculotendon (MT) mechanics during treadmill running at varying speeds. **Methods**: Eighteen healthy young adults ran on a treadmill at 70% (5.21 ± 0.62 m/s), 80% (5.96 ± 0.71 m/s), 85% (6.33 ± 0.76 m/s), 90% (6.70 ± 0.80 m/s), 95% (7.07 ± 0.84 m/s), and 100% (7.44 ± 0.89 m/s) of their maximal sprinting speed. Kinematic data were simultaneously collected using both an optical motion capture (OMC) system (Vicon) and an IMU system (Xsens), while electromyographic (EMG) data recorded hamstring activity. MSK modelling was applied to both kinematic measurements to calculate lower-limb joint moments and hamstring MT mechanics, with estimated muscle activations validated against the EMG data. **Results**: IMU-based estimations closely matched OMC-based calculations, with coefficient of multiple correlations (CMC) exceeding 0.85 for hip and knee joint moments during swing and 0.95 for hamstring MT kinematics across full stride cycles at all speeds. MT force estimations varied among hamstring muscles, with semimembranosus showing the highest agreement ($0.96 < CMC < 0.98$) across all speeds. Linear mixed models showed for each 1 m/s speed increase, root mean square errors between the two systems increased by less than 0.25 N/m for joint moments and 0.05 BW for hamstring MT forces. **Conclusions**: IMU-MSK integration is a valid alternative to OMC for estimating sagittal-plane joint moments and hamstring MT mechanics during treadmill running, though differences in peak hip moment during terminal swing warrant caution in field-based applications.

Key Words: IMU, RUNNING MECHANICS, VALIDATION, BLAND-ALTMAN ANALYSIS, BIOMECHANICAL MODEL

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

INTRODUCTION

Accurately assessing joint kinetics and muscle mechanics during dynamic activities such as running is essential for understanding human movement biomechanics, preventing injuries, and enhancing athletic performance. The integration of musculoskeletal (MSK) modelling with biomechanical measurements from optical motion capture (OMC) systems has been widely employed to quantify joint and muscle mechanics across various locomotor tasks (1-4). This integration is widely considered as the reference method for analysing human movement due to the high accuracy provided by OMC systems (5). However, OMC necessitates a controlled laboratory environment and costly equipment, limiting its feasibility for field-based assessments.

The recent advent of wearable inertial measurement units (IMUs) offers a promising and relatively inexpensive alternative to OMC. IMU systems have proven to be practical and cost-effective for directly capturing the accelerations and dynamics of limbs during sprinting, making them a prominent tool for collecting on-field kinematic data across various activities (6, 7). For instance, recent studies validated IMU-based measurements against OMC-based values, demonstrating that IMU systems are valid instruments for measuring lower-limb joint flexion-extension during sprinting at speeds greater than 6 m/s with a root mean square error (RMSE) of less than 10° (8, 9).

While IMU-based joint kinematics estimation has been extensively studied, fewer investigations have validated IMU-based joint kinetics against concurrent OMC measurements using either inverse dynamics or machine learning methods (10). The traditional inverse dynamics approach, which calculates net joint moments by integrating a biomechanical model with experimental measurements of joint kinematics and external forces (e.g., ground reaction forces), remains a common method (11). Recent advancements in deep learning, particularly neural networks, have made machine learning methods

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

increasingly attractive for estimating joint kinetics from IMU data (10, 12). Additionally, optimal control methods have shown promise in accurately predicting joint kinetics from IMU data. For instance, Dorschky et al. (13) solved an optimal control problem by generating a dynamically consistent simulation of running directly from raw IMU data. Both machine learning and optimal control methods offer the advantage of estimating joint kinetics in field-based settings where external force measurements are impractical. However, these two approaches typically require longer preparation time and greater computational resources compared to the inverse-dynamics method.

Employing the well-established and computationally efficient inverse dynamics approach, the aim of the present validation study was twofold. First, to validate the integration of MSK modelling with IMU measurements for calculating sagittal-plane joint moments during the swing phase of treadmill running across multiple speeds. Based on previous findings of greater discrepancies in hip compared to knee flexion angles using IMUs during sprinting (8), we hypothesise that IMU-based hip flexion moments will exhibit greater differences than knee flexion moments when compared with OMC-based results.

To broaden the impact of validating the IMU-MSK integration during swing, the second goal of our study is to enhance the understanding of hamstring musculotendon (MT) mechanics over a range of running speeds. Previous studies revealed that hamstring injuries predominantly occur during the late swing phase of high-speed running, as the hamstrings undergo eccentric contraction to decelerate the leg and prepare for foot strike (14, 15). While OMC-MSK integration has been employed to study hamstring MT mechanics during running (15, 16), the validity of IMU-MSK integration for this purpose remains largely unexplored. To fill this gap, our second aim is to establish the validity of IMU-MSK integration in calculating hamstring MT forces during the swing phase of running at different speeds, as well as examining hamstring MT length changes across the entire stride cycle. We

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

hypothesise that the hamstring MT mechanics derived from IMU-MSK integration will demonstrate at least moderate waveform agreement with those obtained from OMC-MSK integration (see Methods section for a detailed definition of 'moderate correlation').

MATERIALS AND METHODS

Participants

Eighteen healthy young adults participated in this study (13 men, 5 women; age: 28.5 ± 5.6 years; height: 175.1 ± 8.3 cm; mass: 74.9 ± 12.2 kg; BMI: 24.2 ± 2.1 kg/m^2). All participants were recreationally active (17) and free from any musculoskeletal injuries that could affect their running performance across the tested speeds. Each participant provided informed consent prior to the commencement of the study, which was approved by the Human Research Ethics Committee.

Data Collection

All participants were required to complete one familiarization session, followed by one testing session, both in the same laboratory. The familiarisation session allowed participants to become comfortable running on an instrumented dual-belt treadmill (FIT, Bertec, Columbus, OH, USA) as well as to determine their maximum speeds. Participants began with walking and slow jogging (1.0 to 3.0 m/s) at speeds set by trained operators. Once comfortable, participants were fitted with a safety harness and tested its fall prevention capability. Wearing the harness, participants then performed a continuous running bout with incrementally increasing speeds until reaching and maintaining at 80% of their perceived maximum speed. Following a minimum of two-minutes rest, participants attempted to reach their perceived maximum speed. If they maintained at least ten strides comfortably at this speed, they rested for another two minutes before attempting a faster speed, increased by

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

either 0.5 m/s or 0.25 m/s, according to their preference. This iterative process continued until participants failed to maintain 10 strides, with their maximum speed defined as the last successful increment. The Bertec Treadmill Control Panel software controlled and recorded treadmill speeds. To ensure consistency, the same pair of trained operators conducted all experiments.

The testing session took place on the same treadmill approximately 13 ± 4 weeks after the familiarisation session. Prior to data collection, participants began a warm-up routine, starting with slow speed running (1.0 to 4.0 m/s). After a brief rest, they completed the warm-up by running at 80% of their maximum speed determined during the familiarisation session. After resting for at least two minutes, the iterative process used in the familiarization session was replicated in each testing session to determine the most recent maximum speed for each participant. Following approximately 30 minutes of rest, motion data acquisition commenced with a 10-second stationary stand in the anatomical position to capture anthropometric measurements for personalised musculoskeletal models. Participants then performed a 'stomp' with a well-defined peak knee flexion to generate a timestamp crucial for later synchronization between two systems (8). Subsequently, participants completed a series of running trials at increasing speeds of 70%, 80%, 85%, 90%, 95%, and 100% of their maximal speed determined at the beginning of the testing session. Achievement of the six target speeds was confirmed verbally by the treadmill operators, after which participants maintained the designated pace for at least 10 consecutive strides (five per leg). Once operators verbally confirmed the 10-stride achievement, participants could stop running and dismount the treadmill. To limit fatigue, a minimum of two minutes of rest was provided between trials.

For each running trial, joint motion, ground reaction forces, and muscle activity were captured simultaneously. Kinematic data were collected at 200 Hz using a 9-camera OMC system (Vicon, Oxford Metrics, Oxford, UK) and at 100 Hz with a seven-sensor IMU system

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

(Xsens MVN Awinda, Movella, Enschede, The Netherlands). Detailed descriptions of retroreflective markers, IMU placements, and system calibration can be found in our previous work (8). Ground reaction forces were measured at 1000 Hz by force plates embedded in the treadmill belt to identify the swing phase for each stride cycle. Muscle activity was recorded using a wireless EMG system (Cometa Miniwave Infinity, Cometa Scientific Ltd., Milan, Italy) at 1000 Hz. Following Surface Electromyography for the Non-Invasive Assessment of Muscles (SENIAM) guidelines, EMG electrodes were placed on the medial and lateral hamstrings. The EMG data underwent high-pass filtering (4th order Butterworth, 20Hz cutoff), full-wave rectification, and low-pass filtering (4th order Butterworth, 8Hz cutoff) to obtain a linear envelope (18). Note that only the EMG data collected from the medial and lateral hamstrings were presented below as these were the muscles of interest.

Muscle Force Calculations

An inverse-dynamics approach was employed in OpenSim (version 4.2) (19) to compute muscle forces during the swing phase of running (1). For each participant, two participant-specific musculoskeletal models were generated: an OMC-based model and an IMU-based model, both derived from a generic full-body model (19). The OMC-based model was created using OpenSim's scale tool to scale the generic model based on marker-based anthropometric measurements collected during each participant's 10-second stationary stand. The IMU-based model was created by firstly obtaining body dimensions estimated by Xsens MVN Analyze 2022.0 (Movella, Enschede, The Netherlands) after providing the subject's body height and foot length measurements. These estimated body dimensions were then incorporated into the generic model via a custom Matlab script (R2022a, The Mathworks, Natick, MA). Each participant-specific model has 10 degree-of-freedom (DOF) legs (hip: 3 DOF, knee: 1 DOF, ankle: 1 DOF) and both legs were actuated by 80 Hill-type MT units (20).

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Lower-limb joint angles and MT kinematics were calculated across the entire stride cycle for both OMC-based and IMU-based models. For OMC-based models, the inverse kinematic algorithm within OpenSim was employed to compute joint angles across all running trials (21). Conversely, joint angles for the IMU-based models were directly retrieved from Xsens MVN Analyse software using the proprietary Kalman Filter (22, 23). A fourth-order Butterworth low-pass filter was used to filter with the raw joint angles obtained from both systems at a cutoff frequency of 8 Hz determined by conducting a residual analysis (24, 25). These filtered joint angles were then input to the corresponding participant-specific model to compute the length and velocity of each MT unit using the "Muscle Analysis" tool available in OpenSim. MT strain for each muscle was calculated as the percentage change in length relative to the reference MT length obtained from setting all lower-limb joint angles to zero.

Net joint moments and MT forces during the swing phase of running were computed for each trial. The net joint moments exerted about the lower-limb joints were computed using standard inverse dynamics analysis for both IMU and MOC measurements (25). These net joint moments were then decomposed into individual MT forces by minimizing the sum of the squares of all muscle activations subject to the force-length and force-velocity properties of the muscles (26). Because the IMU-based pelvis position was fixed at the OpenSim world frame origin throughout the stride, experimental force plate data could not be directly applied to the IMU-based models for computing net joint moments during stance. Consequently, comparisons of net joint moments and MT forces between the OMC and IMU systems were limited to the swing phase, despite the availability of force-plate data for the entire stride cycle. All joint moments and MT forces were normalised by each participant's body mass (BM) and body weight (BW), respectively. The muscles of interest for further analysis were biceps femoris long head (BFLH), biceps femoris short head (BFSH), semimembranosus (SM), and semitendinosus (ST).

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Data Analysis

The OMC-based and IMU-based data were synchronized before the comparison between the two systems. The synchronisation was performed by using the timestamps of the peak knee flexion angle obtained from the two systems during the stomp. Subsequently, ten continuous strides were identified for each trial. Each stride cycle began and ended with the ipsilateral foot-strike, while each swing cycle began with the ipsilateral toe-off and ended with the subsequent ipsilateral foot-strike. Foot-strike and toe-off events were identified using force plates embedded in the treadmill, with a threshold of 10% body weight (BW) applied to the vertical ground reaction force to detect these events (27). To assess synchronization performance, we evaluated the agreement of post-synchronization peak knee flexion times between the two systems. Specifically, the intraclass correlation coefficient (ICC) was calculated to measure absolute agreement across ten consecutive stride cycles for each participant at each running speed. The analysis demonstrated excellent temporal alignment between the two systems, with all ICC values exceeding 0.99.

To quantitatively assess the differences between the OMC-based and IMU-based data for each stride cycle, root mean square error (RMSE), normalised RMSE (NRMSE), and coefficient of multiple correlation (CMC) were calculated for the measures of interest. Specifically, we computed these three continuous metrics for sagittal-plane joint angles and MT kinematics (strains and velocities) using complete stride cycles. However, for sagittal-plane joint moments and MT forces, these metrics were calculated using only the swing phase data from each stride cycle. The NRMSE was computed by normalising the RMSE with a range derived by averaging peak-to-peak values from both systems (28). CMC provides a comprehensive assessment of overall waveform agreement that accounts for both pattern and magnitude similarities across multiple stride cycles (29). According to Ferrari et al. (30), CMC values were interpreted as follows: < 0.65 indicated weak agreement, 0.65–

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

0.75 indicated moderate agreement, 0.75–0.85 indicated good agreement, 0.85–0.95 indicated very good agreement, and ≥ 0.95 indicates excellent agreement.

Linear mixed models (LMMs) were used to assess how running speed affects RMSE values, as they effectively handle hierarchical data structures and within-subject correlations (31). For each sagittal plane joint angle and MT kinematic, RMSE was calculated per trial using data from 10 continuous stride cycles. The RMSE for each sagittal plane joint moment and MT force, however, was calculated based on swing phase data from 10 continuous stride cycles. Separate LMMs were fitted with each type of RMSE as the dependent variable, running speed as a fixed factor, and participant ID as a random factor. Each LMM was formulated as:

$$\boldsymbol{y} = \beta_0 + \boldsymbol{X}\beta + \boldsymbol{u} + \boldsymbol{\epsilon}$$

where $\mathbf{y}$ is the vector of RMSE values, β_0 is the fixed intercept, $\boldsymbol{X}$ is the vector of running speed, β is the fixed effect coefficient (or slope), $\boldsymbol{u}$ is the vector of random effect for participant, and $\boldsymbol{\epsilon}$ is the vector of residual error. The slope represents the change in RMSE value for a one-unit increase in speed (i.e., 1 m/s). Statistical significance for the intercept and fixed effect coefficient was set at $p < 0.05$.

Multiple discrete metrics were also extracted from both datasets to evaluate agreement between the OMC and IMU systems using Bland–Altman analysis (32). Peak sagittal-plane joint angles from the IMU and OMC systems were identified for all available stride cycles, while peak hip extension and knee flexion moments, MT forces, and MT kinematics were identified during the swing phase of running. Bland–Altman plots were generated separately for each discrete variable to evaluate the IMU system by determining systematic bias and limits of agreement (LoA) (8) using an open-source Bland–Altman and Correlation Plot

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

toolbox (33).

To visualize differences in measures of interest between the OMC and IMU systems, data from both were time normalised. For each system, the sagittal-plane joint angles and MT kinematics were normalised to a 0-100% scale representing the entire stride cycle. Conversely, the sagittal-plane joint moments and MT forces were normalised to a 0-100% scale representing only the swing phase. For each participant, time-normalised data from 10 continuous stride cycles per trial were averaged to generate specific mean curves. These subject mean curves were then averaged across all participants to generate group mean plots with standard deviation (± 1SD).

RESULTS

Data Collection and Running Speeds

Data collection comprised 2160 total possible stride cycles from 18 participants performing running trials across six distinct speeds (18 participants × 6 speeds × 2 trials × 10 cycles). Six trials were excluded due to missing marker data, leaving 2100 stride cycles for analysis. The average peak running speeds for 70%, 80%, 85%, 90%, 95%, and 100% of maximum speed were 5.21 ± 0.62 m/s (range: 4.20-6.30 m/s), 5.96 ± 0.71 m/s (range: 4.80-7.20 m/s), 6.33 ± 0.76 m/s (range: 5.10-7.65 m/s), 6.70 ± 0.80 m/s (range: 5.40-8.10 m/s), 7.07 ± 0.84 m/s (range: 5.70-8.55 m/s), and 7.44 ± 0.89 m/s (range: 6.00-9.00 m/s), respectively.

Continuous Measurements of the Lower-Limb Joint Angles and Moments

Excellent agreement was observed between the IMU-based and OMC-based lower-limb joint angle patterns across all running speeds, with the ankle joint at maximal speed showing the lowest median CMC value of 0.97 (Supplemental Figure 1, which compares IMU-based

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

joint angles with OMC-based results, Supplemental Digital Content). The median RMSE value remained below 10° for the hip, knee, and ankle joint angles at all running speeds. Specifically, the median RMSE values increased from 7.8° to 9.1°, 4.8° to 6.2°, and 4.6° to 6.6°, for the hip, knee, and ankle joint angles, respectively, as the running speed increased from 70% to the maximal. When considering the range of motion, the knee joint exhibited the lowest median NRMSE values, ranging from 3.9% to 4.9% across all speeds (Supplemental Figure 2, which presents the validity of IMU-based joint angles using continuous metrics, Supplemental Digital Content).

The IMU system accurately reproduced the general patterns of OMC-based hip and knee joint moments during the running swing phase (Figure 1), with excellent agreement evidenced by median CMC values greater than 0.96 (Supplemental Figure 2, which presents the validity of IMU-based joint moments using continuous metrics, Supplemental Digital Content). While the median RMSE values for knee joint moment remained below 0.30 Nm/kg across all running speeds, they ranged from 0.67 Nm/kg at 70% of the maximal speed to 0.85 Nm/kg at the 95% of maximal speed for the hip joint moment. Additionally, the knee joint moment displayed lower median NRMSE values, ranging from 10.7% to 12.8%, compared to the hip joint moment, which ranged from 12.4% to 17.2%. The LMM analysis also indicated that for every 1 m/s increase in running speed, the RMSE for hip and knee flexion moments significantly increased by 0.23 Nm/kg ($p < 0.001$) and 0.04 Nm/kg ($p < 0.001$), respectively (Supplemental Table 1, which presents the results of LMM analysis, Supplemental Digital Content).

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Continuous Measurements of the Hamstring Musculotendon Dynamics

The sequence and timing of the predicted activations for BFLH and SM muscles were qualitatively consistent with the measured EMG (Supplemental Figure 3, which compares predicted muscle activations with experimental EMG data, Supplemental Digital Content). Both the predicted activations and experimental EMG showed a gradual increase in BFLH and SM activity starting at approximately 60% of the swing phase, with peak activation occurring at the terminal swing across all walking speeds.

The waveform agreement between the OMC- and IMU-based muscle force solutions during swing ranged from good to excellent across four hamstring muscles (Figure 2), with median CMC values highest for SM force at maximal speed (0.98) and lowest for BFSH force at 70% speed (0.88) (Supplemental Figure 4, which presents the validity of IMU-based hamstring MT mechanics using continuous metrics, Supplemental Digital Content). The median RMSE for the BFLH and SM forces were approximately three times higher (ranging from 0.32 BW to 0.39 BW across all speeds) than that for the BFSH and ST forces (ranging from 0.06 BW to 0.15 BW across all speeds). Peak SM force occurrences were consistent between OMC and IMU solutions, while the BFLH force peaked earlier in the OMC solution compared to the IMU solution at all speeds (Figure 2). Additionally, running speed significantly influenced the difference between OMC- and IMU-based muscles force for all four hamstring muscles, with RMSE for BFLH and SM increasing by 0.01 BW ($p < 0.01$) and 0.04 BW ($p < 0.001$), respectively, for every 1 m/s increase in running speed (Supplemental Table 1, which presents the results of LMM analysis, Supplemental Digital Content).

The IMU-based MT strain and velocity for hamstring muscles were in excellent agreement with the OMC-based estimations with the median CMC values greater than 0.98 regardless of the speed (Figures 3–4 and Supplemental Figures 5 and 6, which present the

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

validity of IMU-based hamstring MT kinematics using continuous metrics, Supplemental Digital Content). The median RMSE and NRMSE of MT strain were less than 3% and 10% across four hamstring muscles and all speeds, respectively. Similarly, all hamstring muscles exhibited lower than 10% NRMSE for their MT velocity across all speeds in which the uniarticular BFSH exhibited the lowest RMSE (ranging from 0.05 m/s to 0.07 m/s) and NRMSE (ranging from 4.8% to 6.6%) compared to the three biarticular hamstring muscles. Additionally, the LMM analysis indicated a 1 m/s speed increase would result in statistically significant increases in the RMSE of hamstring MT strain (ranging from 0.02% for BFSH to 0.07% for BFLH; $p < 0.001$) and hamstring MT velocity (ranging from 0.01 m/s for BFSH to 0.02 m/s for ST; $p < 0.001$) (Supplemental Table 1, which presents the results of LMM analysis, Supplemental Digital Content). However, the practical impact of these increases was limited when compared to the corresponding peak-to-peak values across one stride cycle. For example, a per-unit-speed increase of 0.02 m/s in the RMSE of MT velocity for ST was relatively inconsequential compared to a magnitude of 1.4 m/s peak-to-peak at 70% speed (Figure 4).

Discrete Measurements of the Lower-Limb Joint Angles and Moments

Bland–Altman analysis revealed that the peak knee flexion angle exhibited the smallest and insignificant systematic bias between the OMC and IMU systems at each running speed (ranging from -0.02° to 0.66° with $p > 0.05$), in contrast to the peak hip (ranging from 1.0° to 3.9° with $p < 0.05$) and ankle (ranging from -0.64° to 3.3° with $p < 0.001$) flexion angles (Supplemental Figure 7, which presents the agreement of IMU-based peak joint angles using Bland-Altman analysis, Supplemental Digital Content). The LoA for the peak hip flexion angle ranged from 14.4° to 18.2° across all speeds, which were approximately twice as wide as those for the peak knee flexion angle (ranging from 7.6° to 9.5°) and the peak ankle

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

flexion angle (ranging from 7.2° to 10.1°).

Across all speeds, significant biases were observed in the peak hip extension moment and peak knee flexion moment ($p < 0.001$), except for the hip ($p = 0.55$) and knee ($p = 0.14$) joints at 70% speed (Figure 5). Specifically, the IMU system consistently overestimated the peak hip extension moment with the significant bias increasing from 0.12 Nm/kg at 80% speed to 0.47 Nm/kg at the maximal speed. Similarly, the IMU system consistently overestimated the peak knee flexion moment with close-to-zero significant biases. As the running speed increased from 80% to the maximal, the systematic bias for the peak knee flexion moment increased from 0.07 Nm/kg to 0.15 Nm/kg.

At 90% of maximal running speed, we observed the highest frequency of hip (n=18) and knee (n=30) moment outliers exceeding the upper LoA. Notably, three participants were responsible for all the knee moment outliers, with two of these participants also contributing all hip moment outliers.

Discrete Measurements of the Hamstring Musculotendon Dynamics

The IMU system underestimated and overestimated the peak MT force for the lateral (BFLH and BFSH) and medial (SM and ST) hamstrings, respectively, across all speeds ≥80% of the maximal running speed (Figure 6). BFLH and SM demonstrated a wider LoA (>0.57 BW) across all speeds compared to BFSH and ST (>0.16 BW). At least 94% of the peak force measurements for BFLH and SM fell within their LoA at all speeds. In contrast, a minimum of 84% and 87% of the peak force measurements for BFSH and ST, respectively, fell within their LoA.

Overall, the IMU system significantly underestimated the peak MT strain for the medial hamstrings, with the bias ranging from 0.45% to 0.88%, irrespective of the running speed (Figure 7). In comparison, the biases for the lateral hamstrings were closer to zero, with

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

BFLH exhibiting the largest bias of 0.51% ($p < 0.01$) at the maximal speed. BFSH demonstrated the narrowest LoA (< 2.1%) across all speeds compared to BFLH (< 5.4%), SM (< 5.2%), and ST (< 6%). For BFLH, SM, and ST, at least 95% of the peak MT strain measurements fell within their respective LoA at all speeds. Regarding BFSH, 86% to 94% of its peak MT strain measurements fell within its LoA across all running speeds.

When examining all speeds, the IMU system consistently underestimated the peak MT lengthening velocity for BFSH with a significant bias that varied between 0.06 m/s ($p < 0.001$) and 0.08 m/s ($p < 0.001$). The IMU system, however, overestimated the peak MT velocity for BFLH, SM, and ST with a significant bias that varied between 0.03 m/s ($p < 0.001$) and 0.11 m/s ($p < 0.001$). BFSH exhibited both the narrowest LoA (ranging from 0.07 m/s to 0.11 m/s) and the maximum number of peak MT velocity measurements within its LoA (ranging from 97% to 99%) when compared to the other three hamstring muscles.

DISCUSSION

The aim of the present study was to validate IMU-MSK integration for estimating sagittal-plane joint moments and hamstring MT mechanics during the treadmill running across multiple speeds. We found that the median RMSE value remained below 10° for the hip, knee, and ankle joint angles at all running speeds. In support of our hypothesis, the IMU-based hip flexion moment exhibited an RMSE approximately 0.5 Nm/kg greater than the knee flexion moments at all running speeds. The IMU system effectively replicated the general patterns of hamstring MT strain and velocity throughout the stride cycle, showing excellent agreements (CMC > 0.95) with the OMC system across all speeds. However, the waveform agreement in MT force between the two systems during the swing phase ranged from very good to excellent (0.88 < CMC < 0.98) across the four hamstring muscles. Consequently, our second hypothesis regarding the agreement in hamstring MT mechanics

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

between the IMU-MSK and OMC-MSK integration was also confirmed.

Our IMU-MSK integration effectively reproduced the overall trends of OMC-based joint moments and MT forces during the swing phase of running. Dorschky et al. (13) compared sagittal-plane joint moments across the entire stride cycle at speeds of 3 m/s to 4.9 m/s, reporting mean RMSE values of 0.57 Nm/kg for the hip and 0.61 Nm/kg for the knee between the OMC and IMU systems, with Pearson correlations greater than 0.85 for both joints. In comparison, our results at a comparable speed (70% of maximum speed, 5.21 m/s) demonstrated excellent waveform agreement for both joints (median CMC = 0.96 for hip, CMC = 0.98 for knee) and similar RMSE values (0.69 Nm/kg for hip, 0.23 Nm/kg for knee) (Supplemental Figure 2, which presents the validity of IMU-based joint angles using continuous metrics, Supplemental Digital Content). The focus on the swing phase likely contributed to our lower knee moment RMSE (0.23 Nm/kg) compared to Dorschky et al. (0.61 Nm/kg) as peak knee moments are known to be substantially lower during swing (3).

The present IMU-MSK integration also successfully captured the patterns and magnitudes of hamstring OMC-based MT mechanics throughout the stride cycle. Both OMC and IMU systems indicated peak hamstring MT forces during the terminal swing phase, with the SM consistently generating the greatest force, followed by the BFLH (Figure 2). This aligns with the previous MSK modelling studies (34, 35) and is supported by a magnetic resonance imaging study showing that SM and BFLH are the largest and the second-largest hamstring muscles, respectively, based on their muscle volumes (36). Consistent with the findings from Chumanov et al. (34), both OMC and IMU systems showed minimal variation in peak MT strain during the swing phase across different running speeds (Figure 3). Furthermore, our results agreed with Chumanov et al. (34) and Schache et al. (35) in demonstrating that the BFLH experienced the greatest peak MT strain during terminal swing at all speeds. Additionally, similar to previous research (35, 37), our analysis revealed that the

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

ST exhibited the highest lengthening velocity throughout the swing phase at all speeds (Figure 4).

It is worth highlighting that separate MSK models were developed for each participant: one for the OMC system and another for the IMU system. These distinct models were then integrated into their respective OMC-MSK and IMU-MSK integrations. As a result, the discrepancies observed between the OMC- and IMU-based results for joint moments and MT mechanics stem not only the kinematic differences between the two systems but also the inherent differences between the two models. To quantify the impact of model differences, we solved an inverse dynamics problem for one participant running at maximal speed by pairing their OMC-based MSK model with IMU-based joint kinematics. This analysis revealed a RMSE of 142 Nm for hip flexion moment across five swing phases compared to the reference OMC-based solution, while the IMU-based solution showed a mean RMSE of 183 Nm. The 41 Nm difference suggests that model selection accounts for approximately 22% (41/183) of the RMSE in hip moments. Further investigation involving additional participants and trials is necessary to validate and generalize these findings.

Accurate estimation of joint moments during running is critical for assessing muscle loading and injury risk, particularly during the terminal swing phase. This phase poses the highest risk for hamstring strain injuries due to the rapid deceleration of the swing leg in preparation for foot-strike. While the current IMU-MSK integration using inverse dynamics captures the overall trend of OMC-based joint moments during swing, notable discrepancies persist during terminal swing, especially in hip moment estimation (Figure 1). Bland–Altman analysis further revealed that peak hip moments during this phase showed limits of agreement up to approximately 50% of peak values (Figure 5). Although some of these differences may stem from discrepancies between the two modelling approaches, further refinement of IMU measurement techniques and validation protocols is necessary to improve accuracy and

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

ensure reliable application in the field.

Both OMC and IMU systems showed that the peak hip and knee moments during the swing phase increased with running speed (Figure 1). OMC-based results revealed a 1.73-fold increase in peak hip extension moment and a 1.62-fold increase in knee flexion moment as speed increased from 70% to 100% of maximal. While the IMU system tracked similar trends, it yielded slightly higher fold-changes for both peak hip (1.97-fold) and knee (1.73-fold) moments. This difference is further highlighted by Bland-Altman analyses, revealing a systematic overestimation of peak moments by the IMU system, particularly for hip moment at higher speeds (Figure 5). The bias increased progressively with speed, reaching a maximum of 0.49 Nm/kg for hip extension at 95% of maximal speed and 0.15 Nm/kg for knee flexion at maximal speed. When normalised the systematic biases by corresponding OMC-based peak moments, the relative systematic bias for the hip moment exceeded 10% as running speeds increased from 85% to the maximum. Conversely, the relative bias for the knee moment remained below 10% at all speeds. Additionally, the median NRMSE value for the hip moment was approximately 1.2 times greater than that for the knee moment across all speeds (Supplemental Figure 2, which presents the validity of IMU-based joint angles using continuous metrics, Supplemental Digital Content). These combined findings suggest that the IMU system performs better for capturing knee joint moments during the swing phase of running compared to hip moments.

Bland-Altman analysis revealed speed-dependent variations in agreement between IMU and OMC for peak joint moments and hamstring forces. Specifically, the LoA for peak hip and knee moments was smallest at the slowest running speed (hip: 1.1 Nm/kg; knee: 0.44 Nm/kg) and largest at the fastest (hip: 1.5 Nm/kg; knee: 0.68 Nm/kg). This increasing discrepancy may be attributed to the higher inertial loads at faster speeds, which could amplify differences in joint moment estimations due to variations in kinematics and model

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

assumptions between the two systems.

Conversely, the LoA for peak BFLH and SM MT forces was highest at the slowest running speed (BFLH: 1.2 BW; SM: 0.85 BW) and lowest at the fastest (BFLH: 0.57 BW; SM: 0.68 BW). This trend may be explained by more consistent muscle activation patterns and force production at higher speeds. Previous studies have shown that BFLH and SM activation levels during the terminal swing of maximal sprinting can approach or even exceed 100% of those measured during maximum voluntary contraction (38, 39). Given that BFLH and SM are the dominant lateral and medial hamstring muscles, respectively, our static optimization criterion would preferentially recruit these muscles during terminal swing to satisfy joint moment requirements. Consequently, both OMC-MSK and IMU-MSK integrations are expected to produce more consistent hamstring force predictions as running approaches maximal speed.

Our comparison of hamstring MT forces between OMC and IMU systems revealed several key points. First, the IMU system exhibited the lowest NRMSE values (ranging from 12% to 18% across all speeds) and the very good correlations (ranging from 0.85 to 0.92 across all speeds) for the SM muscle compared to other hamstring muscles (Supplemental Figure 4, which presents the validity of IMU-based hamstring MT mechanics using continuous metrics, Supplemental Digital Content). Second, the IMU system captured the timing of peak SM force more closely to the OMC-based value than it did for peak BFLH force across all running speeds (Figure 2). Specifically, the IMU system tended to predict the occurrence of peak BFLH slightly later than the OMC system, with a median time difference ranging from 0.03 s late at 70% to 0.02 s late at 100% of maximum speed. For peak SM force, these median time differences were reduced to 0.015 s earlier at 70% and zero at 100% of maximum speed (Figure 2). Third, Bland-Altman analysis revealed a lower systematic bias for peak SM force compared to BFLH force at most speeds (Figure 6). Overall, these findings

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

suggest that the current IMU system has promising potential as an OMC alternative for predicting the magnitude and timing of peak hamstring force during sprinting, particularly for SM, the largest medial hamstring muscle.

Both OMC and IMU systems showed that peak hamstring MT strain remained relatively constant across running speeds (Figure 3). Peak strain ranges for BFLH (9.6%-10.2% for OMC, 9.0%-9.8% for IMU), SM (8.4%-8.8% for OMC, 7.2%-7.7% for IMU), and ST (8.7%-9.2% for OMC, 7.7%-8.2% for IMU) showed minimal variation with speed (Figure 3). Notably, Bland-Altman analysis revealed that BFLH, the hamstring most susceptible to injury (40, 41), exhibited the smallest bias (ranging from -0.51% to 0.35%) and comparable LoA (ranging from 4.12% to 5.32%) across all speeds compared to the medial hamstring muscles (Figure 7). When the systematic biases were normalised by the corresponding OMC-based peak MT strain, the relative systematic bias for BFLH remained remarkably low, reaching a maximum of 5% at maximal speed. In contrast, medial hamstring muscles exhibited larger relative systematic biases ranging from 5.1% to 10.2%. These findings highlight the IMU system's ability to effectively track BFLH MT strain, which is valuable for monitoring injury risk in this muscle.

Contrary to the peak MT strain, both OMC and IMU systems demonstrated that the peak hamstring MT lengthening velocity increased gradually with the running speed (Figure 4). As the running speed increased from 70% to 100% of maximum speed, all hamstring muscles exhibited a 1.3-fold increase in their peak MT lengthening velocities, regardless of the system used. This is in line with Thelen et al. (37), who reported a similar 1.1-fold increase for BFLH, SM, and ST muscles when the running speed increased from 80% (7.13 m/s) to 100% of the maximum speed (8.92 m/s). However, Bland-Altman analysis demonstrated a systematic overestimation of peak lengthening velocity by the IMU system for BFLH, SM, and ST muscles at all speeds (Figure 8). BFLH showed the largest bias (ranging from 0.09

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

m/s to 0.11 m/s), while ST exhibited the smallest (ranging from 0.03 m/s to 0.05 m/s). When we normalised the bias by the corresponding OMC-based peak lengthening velocities, the relative bias for ST remained below 10% at all speeds. In contrast, the BFLH displayed a larger relative bias of approximately 16%. These results suggest that while the IMU system is generally effective in estimating hamstring lengthening velocity during running, it exhibits varying levels of agreement across different hamstring muscles.

Running speed had a relatively small but statistically significant effect on the differences between OMC and IMU measurements of hamstring mechanics. Our LMM analysis indicated that with a 1 m/s increase in running speed, the RMSE between OMC- and IMU-based hamstring MT mechanics increased slightly, remaining under 0.04 BW for MT force, 0.08% for MT strain, and 0.02 m/s for MT velocity across all four hamstring muscles (Supplemental Table 1, which presents the results of LMM analysis, Supplemental Digital Content). To better contextualize these RMSEs, we normalised them by the corresponding OMC-based peak values. The resulting relative RMSEs for BFLH and SM were less than 2% across all MT metrics and running speeds. These results suggest that the influence of running speed on MT mechanics is relatively small across the four hamstring muscles.

It's crucial to acknowledge certain limitations of our study when interpreting the findings. First, the present validation results are based on a single IMU system (Xsens) and one OMC system (Vicon), which should not be universally applied to other systems. Additionally, IMU-based joint angles were directly obtained from the proprietary Kalman Filter-based sensor fusion algorithm provided by the manufacturer's software (22, 23). Other sensor fusion methods such as Madgwick algorithm (42) and deep learning (43, 44) have been widely used to obtain IMU-based joint kinematics. While the IMU system and associated software used in the current study is widely recognised for analysing human movement across different daily activities (8, 45, 46), utilising multiple IMU systems would

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

have broadened the conclusions.

Second, we calculated joint moments and muscle forces using inverse dynamics. While this approach is widely used, it has inherent limitations. Specifically, the separate measurement of kinematics and external forces (e.g., motion capture for kinematics and force plates for external forces) can create dynamic inconsistencies in the calculations. Although our focus on the swing phase mitigates errors from force-kinematic inconsistencies, both OMC and IMU measurements remain affected by soft tissue artifacts, which can impact the accuracy of joint moment estimates.

Third, our comparisons of joint moments and hamstring muscle forces were limited to the swing phase. Given the comparable peak muscle forces reported during stance and swing phases (34), future studies should validate IMU systems in estimating hamstring mechanics throughout the entire stride cycle. Additionally, the maximum running speed achieved in our study (7.44 ± 0.89 m/s) may not be reflective of higher running speeds.

In conclusion, our study demonstrates moderate to excellent waveform agreements between inverse dynamics-based OMC-MSK and IMU-MSK integrations in estimating sagittal-plane joint moments and key aspects of hamstring mechanics during treadmill running. We found excellent correlations between the two systems for hip and knee flexion moments during swing across all speeds, with the IMU system providing better estimates of knee moment compared to hip moment. Both IMU-MSK and OMC-MSK integrations yielded similar hamstring MT mechanics, with discrepancies between systems increasing slightly at higher speeds. Bland-Altman analysis revealed that joint moment differences grew with speed, likely due to greater inertial loads, whereas hamstring force estimates became more consistent, possibly reflecting stabilized muscle activation patterns. These findings support the potential of the IMU-MSK integration for assessing swing-phase hamstring MT forces and monitoring full-stride hamstring MT strain in on-field applications. However, the

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

notable discrepancies in peak hip moment estimation during terminal swing highlight the need for caution and further refinement before widespread implementation.

Acknowledgments

The study was funded by the National Football League and the Study Melbourne Research Partnerships Program (Grant Number: 44012). The authors declare that the research was conducted without any commercial or financial relationships that could be seen as potential conflicts of interest. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of this study do not constitute endorsement by the American College of Sports Medicine.

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

REFERENCES

1. Pandy MG, Andriacchi TP. Muscle and joint function in human locomotion. *Annu Rev Biomed Eng*. 2010;12:401-33.
2. Maniar N, Schache AG, Pizzolato C, Opar DA. Muscle contributions to tibiofemoral shear forces and valgus and rotational joint moments during single leg drop landing. *Scand J Med Sci Sports*. 2020;30(9):1664-74.
3. Schache AG, Blanch PD, Dorn TW, Brown NAT, Rosemond D, Pandy MG. Effect of running speed on lower limb joint kinetics. *Med Sci Sports Exerc*. 2011;43(7):1260-71.
4. Lin YC, Fok LA, Schache AG, Pandy MG. Muscle coordination of support, progression and balance during stair ambulation. *J Biomech*. 2015;48(2):340-7.
5. van der Kruk E, Reijne MM. Accuracy of human motion capture systems for sport applications; state-of-the-art review. *Eur J Sport Sci*. 2018;18(6):806-19.
6. Horsley BJ, Tofari PJ, Halson SL, et al. Does site matter? Impact of inertial measurement unit placement on the validity and reliability of stride variables during running: a systematic review and meta-analysis. *Sports Med*. 2021;51(7):1449-89.
7. Zeng ZW, Liu Y, Hu XY, Tang MH, Wang L. Validity and reliability of inertial measurement units on lower extremity kinematics during running: a systematic review and meta-analysis. *Sports Med Open*. 2022;8(1):86.
8. Lin YC, Price K, Carmichael DS, et al. Validity of inertial measurement units to measure lower-limb kinematics and pelvic orientation at submaximal and maximal effort running speeds. *Sensors (Basel)*. 2023;23(23):9599.
9. Wilmes E, de Ruiter CJ, Bastiaansen BJC, et al. Inertial sensor-based motion tracking in football with movement intensity quantification. *Sensors (Basel)*. 2020;20(9):2527.
10. Lee CJ, Lee JK. Inertial motion capture-based wearable systems for estimation of joint kinetics: a systematic review. *Sensors (Basel)*. 2022;22(7):2507.
11. Erdemir A, McLean S, Herzog W, van den Bogert AJ. Model-based estimation of muscle forces exerted during movements. *Clin Biomech (Bristol)*. 2007;22(2):131-54.
12. Moghadam SM, Yeung TD, Choisne J. A comparison of machine learning models' accuracy in predicting lower-limb joints' kinematics, kinetics, and muscle forces from

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

wearable sensors. *Sci Rep*. 2023;13(1):5046.

13. Dorschky E, Nitschke M, Seifer AK, van den Bogert AJ, Eskofier BM. Estimation of gait kinematics and kinetics from inertial sensor data using optimal control of musculoskeletal models. *J Biomech*. 2019;95:109278.

14. Opar DA, Williams MD, Shield AJ. Hamstring strain injuries factors that lead to injury and re-injury. *Sports Med*. 2012;42(3):209-26.

15. Kenneally-Dabrowski CJB, Brown NAT, Lai AKM, Perriman D, Spratford W, Serpell BG. Late swing or early stance? A narrative review of hamstring injury mechanisms during high-speed running. *Scand J Med Sci Sports*. 2019;29(8):1083-91.

16. Thelen DG, Chumanov ES, Sherry MA, Heiderscheit BC. Neuromusculoskeletal models provide insights into the mechanisms and rehabilitation of hamstring strains. *Exerc Sport Sci Rev*. 2006;34(3):135-41.

17. McKay AKA, Stellingwerff T, Smith ES, et al. Defining training and performance caliber: a participant classification framework. *Int J Sport Physiol*. 2022;17(2):317-31.

18. Lai A, Schache AG, Brown NAT, Pandy MG. Human ankle plantar flexor muscle-tendon mechanics and energetics during maximum acceleration sprinting. *J R Soc Interface*. 2016;13(121):20160391.

19. Lai AKM, Arnold AS, Wakeling JM. Why are antagonist muscles co-activated in my simulation? A musculoskeletal model for analysing human locomotor tasks. *Ann Biomed Eng*. 2017;45(12):2762-74.

20. Zajac FE. Muscle and tendon - properties, models, scaling, and application to biomechanics and motor control. *Crit Rev Biomed Eng*. 1989;17(4):359-411.

21. Lu T, O'Connor J. Bone position estimation from skin marker co-ordinates using global optimisation with joint constraints. *J Biomech*. 1999;32(2):129-34.

22. Luinge HJ, Roetenberg D, Slycke PJ. Motion tracking system. *Patent US*. 2008;8165844:B2.

23. Roetenberg D, Luinge H, Slycke P. Xsens MVN: Full 6DOF human motion tracking using miniature inertial sensors. *Xsens Motion Tech BV Tech Rep*. 2009;1:1-7.

24. Lin Y-C, Price K, Carmichael DS, et al. Validity of inertial measurement units to measure lower-limb kinematics and pelvic orientation at submaximal and maximal

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

effort running speeds. *Sensors (Basel)*. 2023;23(23):9599.

25. Winter DA. *Biomechanics and motor control of human movement*. John Wiley & Sons; 2009.

26. Anderson FC, Pandy MG. Static and dynamic optimization solutions for gait are practically equivalent. *J Biomech*. 2001;34(2):153-61.

27. Riley PO, Dicharry J, Franz J, Della Croce U, Wilder RP, Kerrigan DC. A kinematics and kinetic comparison of overground and treadmill running. *Med Sci Sports Exerc*. 2008;40(6):1093-100.

28. Ren L, Jones RK, Howard D. Whole body inverse dynamics over a complete gait cycle based only on measured kinematics. *J Biomech*. 2008;41(12):2750-9.

29. Ferrari A, Cutti AG, Cappello A. A new formulation of the coefficient of multiple correlation to assess the similarity of waveforms measured synchronously by different motion analysis protocols. *Gait Posture*. 2010;31(4):540-2.

30. Ferrari A, Cutti AG, Garofalo P, et al. First in vivo assessment of "Outwalk": a novel protocol for clinical gait analysis based on inertial and magnetic sensors. *Med Biol Eng Comput*. 2010;48(1):1-15.

31. Garson GD. Fundamentals of hierarchical linear and multilevel modeling. SAGE Publications; 2013, pp. 3-26.

32. Bland JM, Altman DG. Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet*. 1986;1(8476):307-10.

33. Klein R [Internet]. MATLAB Central File Exchange. Available from: https://www.mathworks.com/matlabcentral/fileexchange/45049-bland-altman-and-correlation-plot.

34. Chumanov ES, Heiderscheit BC, Thelen DG. Hamstring musculotendon dynamics during stance and swing phases of high-speed running. *Med Sci Sports Exerc*. 2011;43(3):525-32.

35. Schache AG, Dorn TW, Blanch PD, Brown NAT, Pandy MG. Mechanics of the human hamstring muscles during sprinting. *Med Sci Sports Exerc*. 2012;44(4):647-58.

36. Handsfield GG, Meyer CH, Hart JM, Abel MF, Blemker SS. Relationships of 35 lower limb muscles to height and body mass quantified using MRI. *J Biomech*.

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

2014;47(3):631-8.

37. Thelen DG, Chumanov ES, Hoerth DM, et al. Hamstring muscle kinematics during treadmill sprinting. *Med Sci Sports Exerc*. 2005;37(1):108-14.

38. Higashihara A, Ono T, Kubota J, Okuwaki T, Fukubayashi T. Functional differences in the activity of the hamstring muscles with increasing running speed. *J Sport Sci*. 2010;28(10):1085-92.

39. McNally T, Edwards S, Halaki M, O'Dwyer N, Pizzari T, Blyton S. Quantifying demands on the hamstrings during high-speed running: a systematic review and meta-analysis. *Scand J Med Sci Spor*. 2023;33(12):2423-43.

40. Koulouris G, Connell DA, Brukner P, Schneider-Kolsky M. Magnetic resonance imaging parameters for assessing risk of recurrent hamstring injuries in elite athletes. *Am J Sports Med*. 2007;35(9):1500-6.

41. Askling CM, Tengvart M, Saartok T, Thorstensson A. Acute first-time hamstring strains during high-speed running - a longitudinal study including clinical and magnetic resonance imaging findings. *Am J Sports Med*. 2007;35(2):197-206.

42. Madgwick SOH, Harrison AJL, Vaidyanathan R. Estimation of IMU and MARG orientation using a gradient descent algorithm. *IEEE Int Conf Rehabil Robot.* 2011;2011:5975346.

43. Hernandez V, Dadkhah D, Babakeshizadeh V, Kulic D. Lower body kinematics estimation from wearable sensors for walking and running: a deep learning approach. *Gait Posture*. 2021;83:185-93.

44. Rapp E, Shin S, Thomsen W, Ferber R, Halilaj E. Estimation of kinematics from inertial measurement units using a combined deep learning and optimization framework. *J Biomech*. 2021;116:110229.

45. Nijmeijer EM, Heuvelmans P, Bolt R, Gokeler A, Otten E, Benjaminse A. Concurrent validation of the Xsens IMU system of lower-body kinematics in jump-landing and change-of-direction tasks. *J Biomech*. 2023;154:111637.

46. Robert-Lachaine X, Mecheri H, Larue C, Plamondon A. Validation of inertial measurement units with an optoelectronic system for whole-body motion analysis. *Med Biol Eng Comput*. 2017;55(4):609-19.

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

FIGURE LEGENDS

Figure 1: Comparison of the sagittal-plane hip and knee joint moments during the swing phase, between IMU-based (red) and OMC-based (blue) measurements, at different running speeds (70%, 80%, 85%, 90%, 95%, and 100% of maximal running speed). Positive moments represent hip extension and knee flexion. The shaded areas (red and blue) represent ± 1 standard deviation from the mean values.

Figure 2: Comparison of the hamstring musculotendon forces during the swing phase, between IMU-based (red) and OMC-based (blue) measurements, at different running speeds (70%, 80%, 85%, 90%, 95%, and 100% of maximal running speed). The shaded areas (red and blue) represent ± 1 standard deviation from the mean values. Muscle symbols are as follows: BFLH, biceps femoris long head; BFSH, biceps femoris short head; SM, semimembranosus; ST, semitendinosus.

Figure 3: Comparison of the hamstring musculotendon strain across one stride cycle, between IMU-based (red) and OMC-based (blue) measurements, at different running speeds (70%, 80%, 85%, 90%, 95%, and 100% of maximal running speed). The shaded areas (red and blue) represent ± 1 standard deviation from the mean values. Vertical dashed line indicates toe-off. Muscle symbols are as follows: BFLH, biceps femoris long head; BFSH, biceps femoris short head; SM, semimembranosus; ST, semitendinosus.

Figure 4: Comparison of the hamstring musculotendon velocity across one stride cycle, between IMU-based (red) and OMC-based (blue) measurements, at different running speeds (70%, 80%, 85%, 90%, 95%, and 100% of maximal running speed). The shaded areas (red

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

and blue) represent ± 1 standard deviation from the mean values. Vertical dashed line indicates toe-off. Muscle symbols are as follows: BFLH, biceps femoris long head; BFSH, biceps femoris short head; SM, semimembranosus; ST, semitendinosus.

Figure 5: Bland-Altman plots illustrate the peak hip extension and knee flexion moments during the swing phase of running at six different speeds. Each square represents the average peak moment on the horizontal axis and the difference between the IMU and OMC systems on the vertical axis. The systematic bias (black horizontal line) and the 95% limits of agreement (black dotted lines) are shown in each plot. Data points outside the limits of agreement are represented by red squares. LoA = Limits of Agreement.

Figure 6: Bland-Altman plots illustrate the peak hamstring musculotendon forces during the swing phase of running at six different speeds. Each square represents the average peak musculotendon force on the horizontal axis and the differences between the IMU and OMC systems on the vertical axis. The systematic bias (black horizontal line) and the 95% limits of agreement (black dotted lines) are shown in each plot. Data points outside the limits of agreement are represented by red squares. LoA = Limits of Agreement.

Figure 7: Bland-Altman plots illustrate the peak hamstring musculotendon strains during running at six different speeds. Each square represents the average peak musculotendon strain on the horizontal axis and the difference between the IMU and OMC systems on the vertical axis. The systematic bias (black horizontal line) and the 95% limits of agreement (black dotted lines) are shown in each plot. Data points outside the limits of agreement are represented by red squares. LoA = Limits of Agreement.

Figure 8: Bland-Altman plots illustrate the peak hamstring musculotendon velocities during

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

running at six different speeds. Each square represents the average peak musculotendon velocity on the horizontal axis and the difference between the IMU and OMC systems on the vertical axis. The systematic bias (black horizontal line) and the 95% limits of agreement (black dotted lines) are shown in each plot. Data points outside the limits of agreement are represented by red squares. LoA = Limits of Agreement.

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

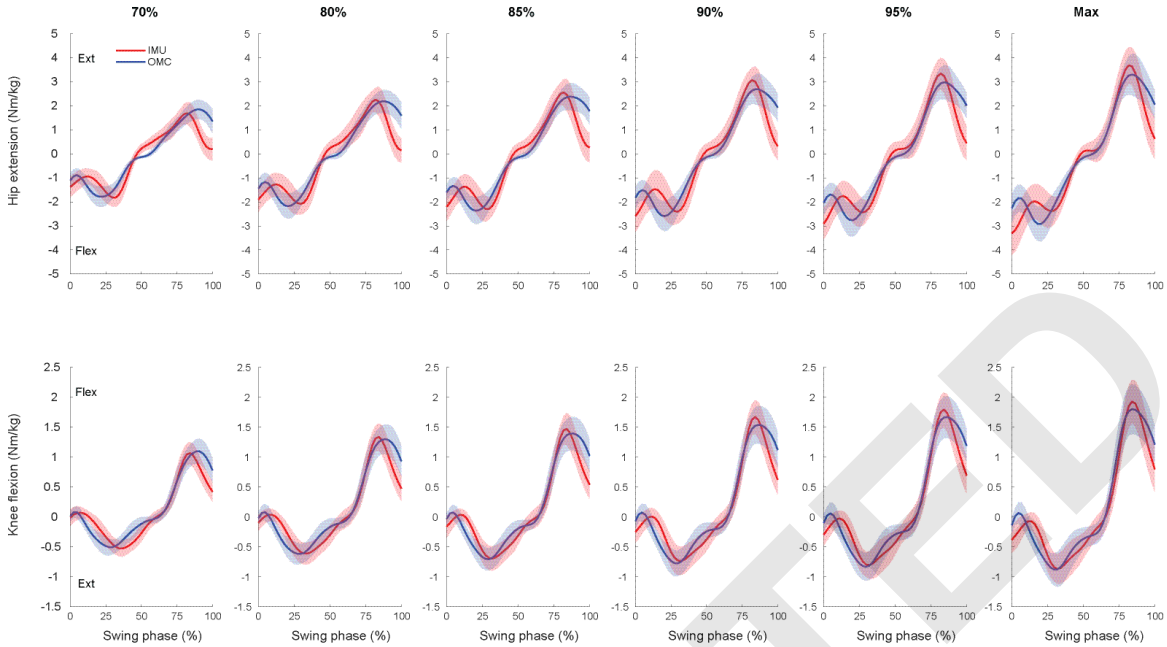
SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Supplemental Digital Content.docx

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

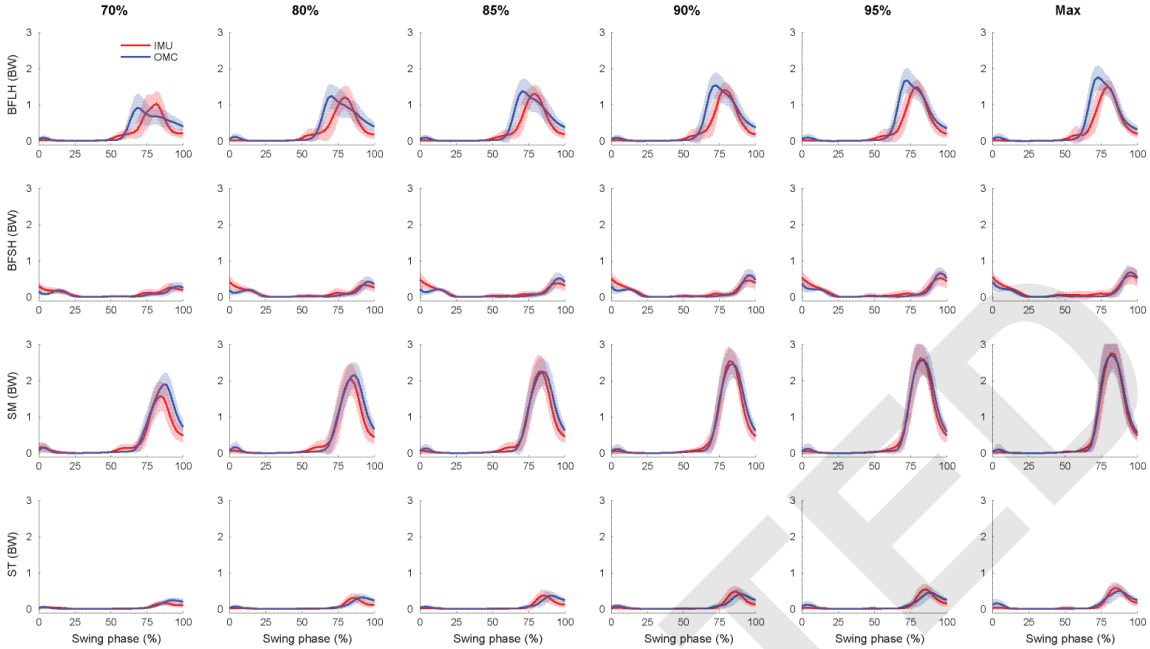
Figure 1



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

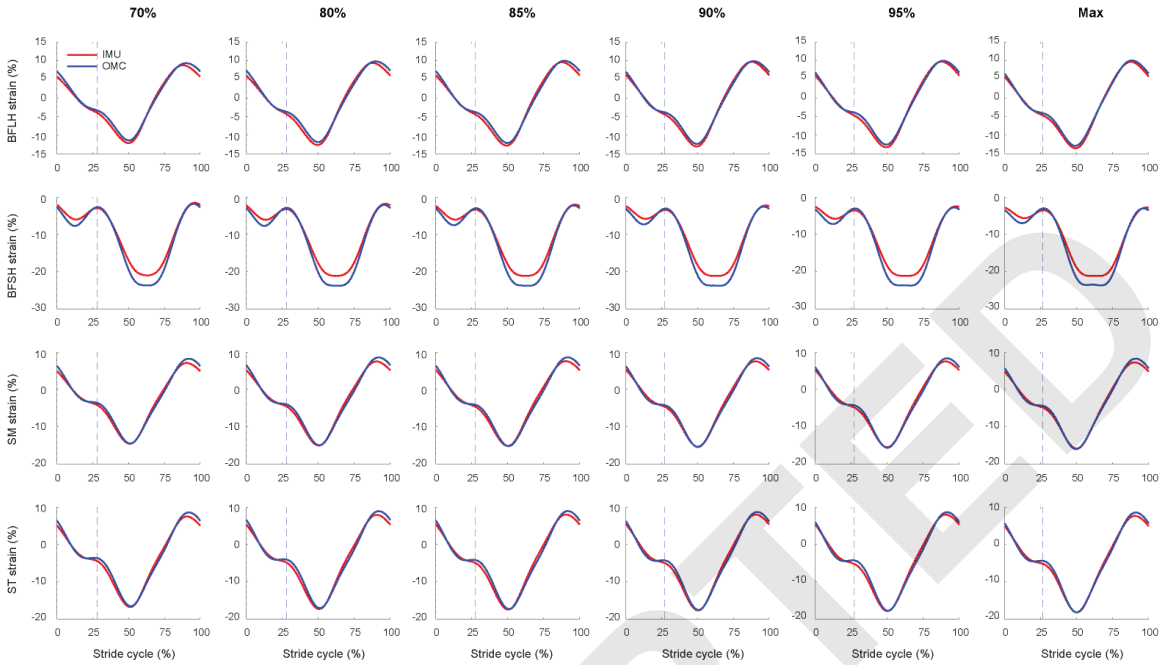
Figure 2



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

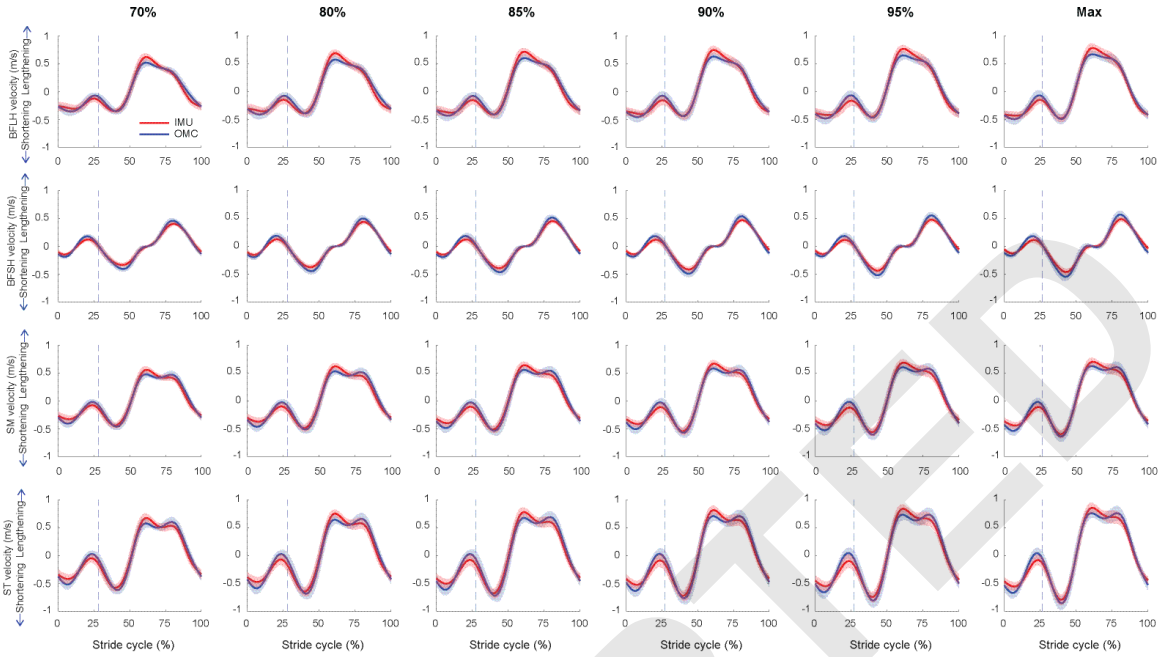
Figure 3



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

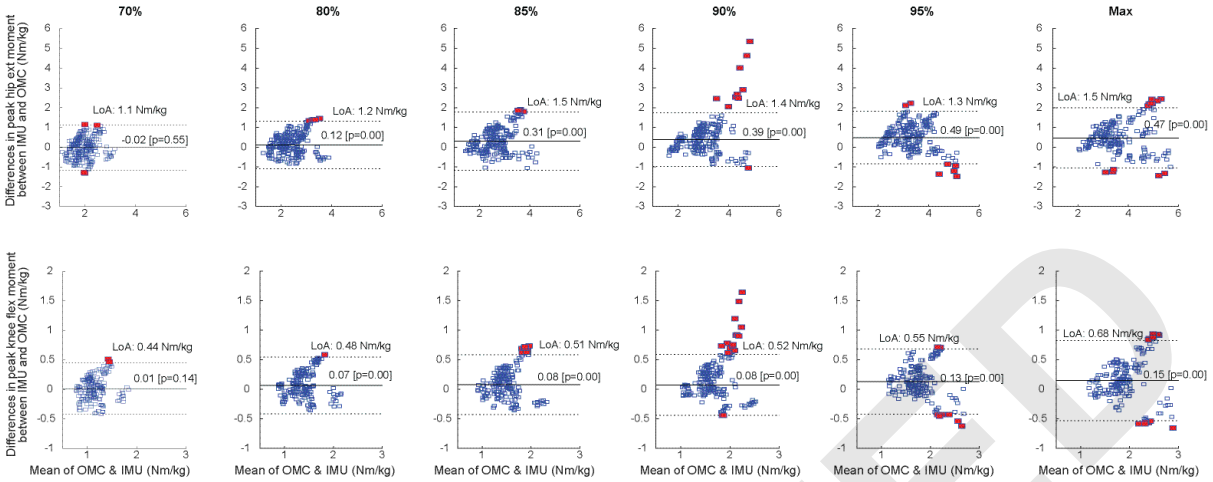
Figure 4



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

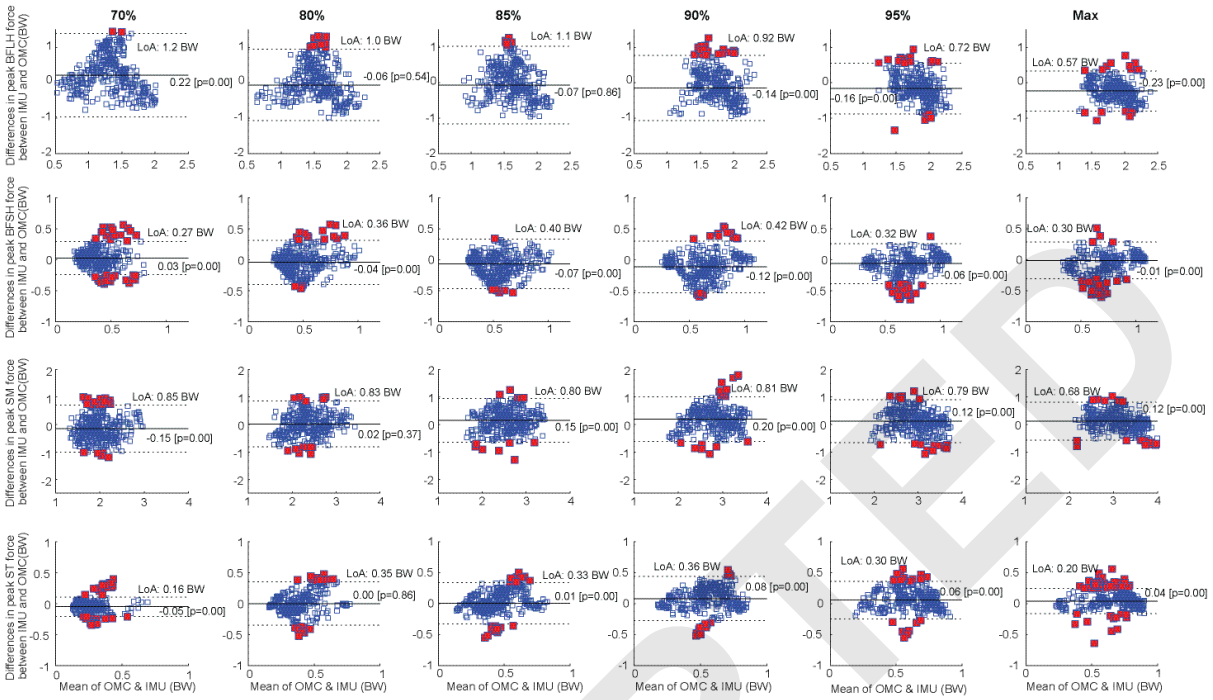
Figure 5



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

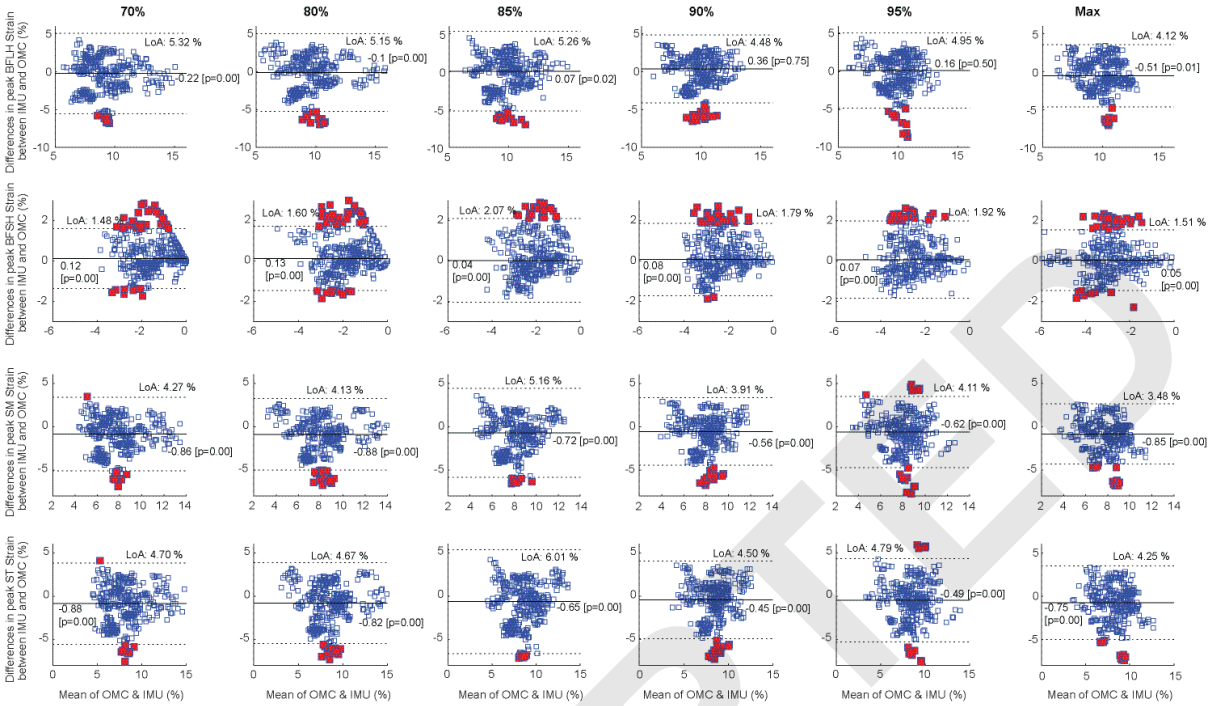
Figure 6



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

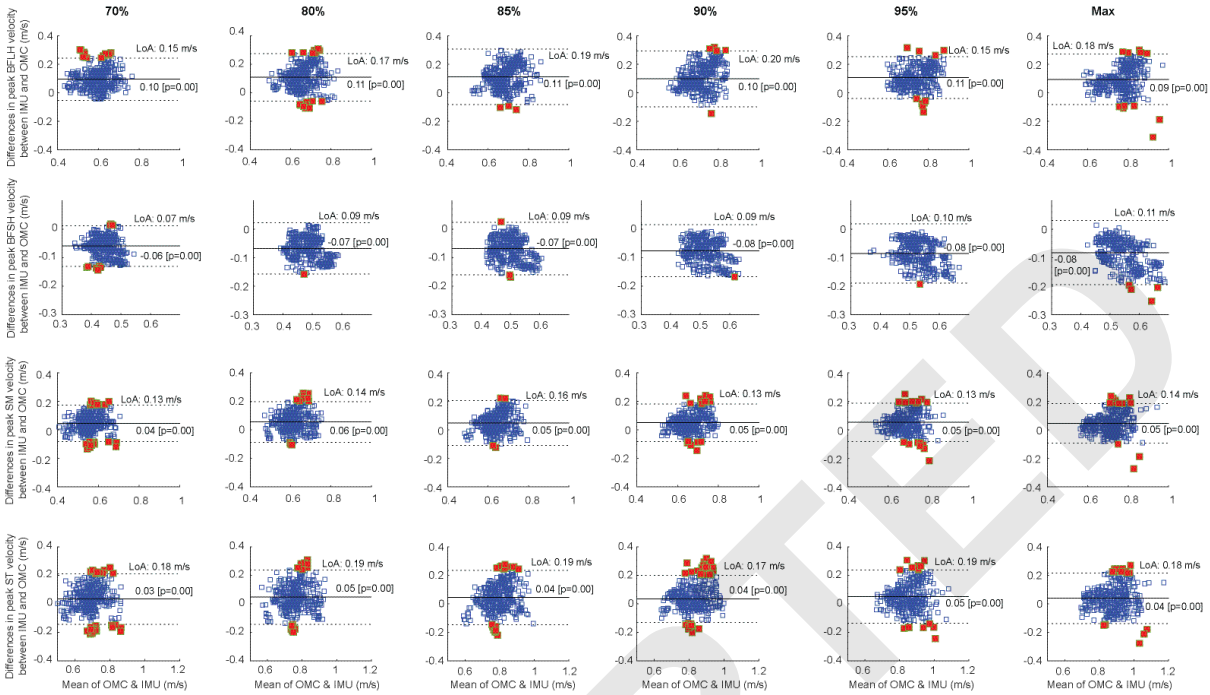
Figure 7



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

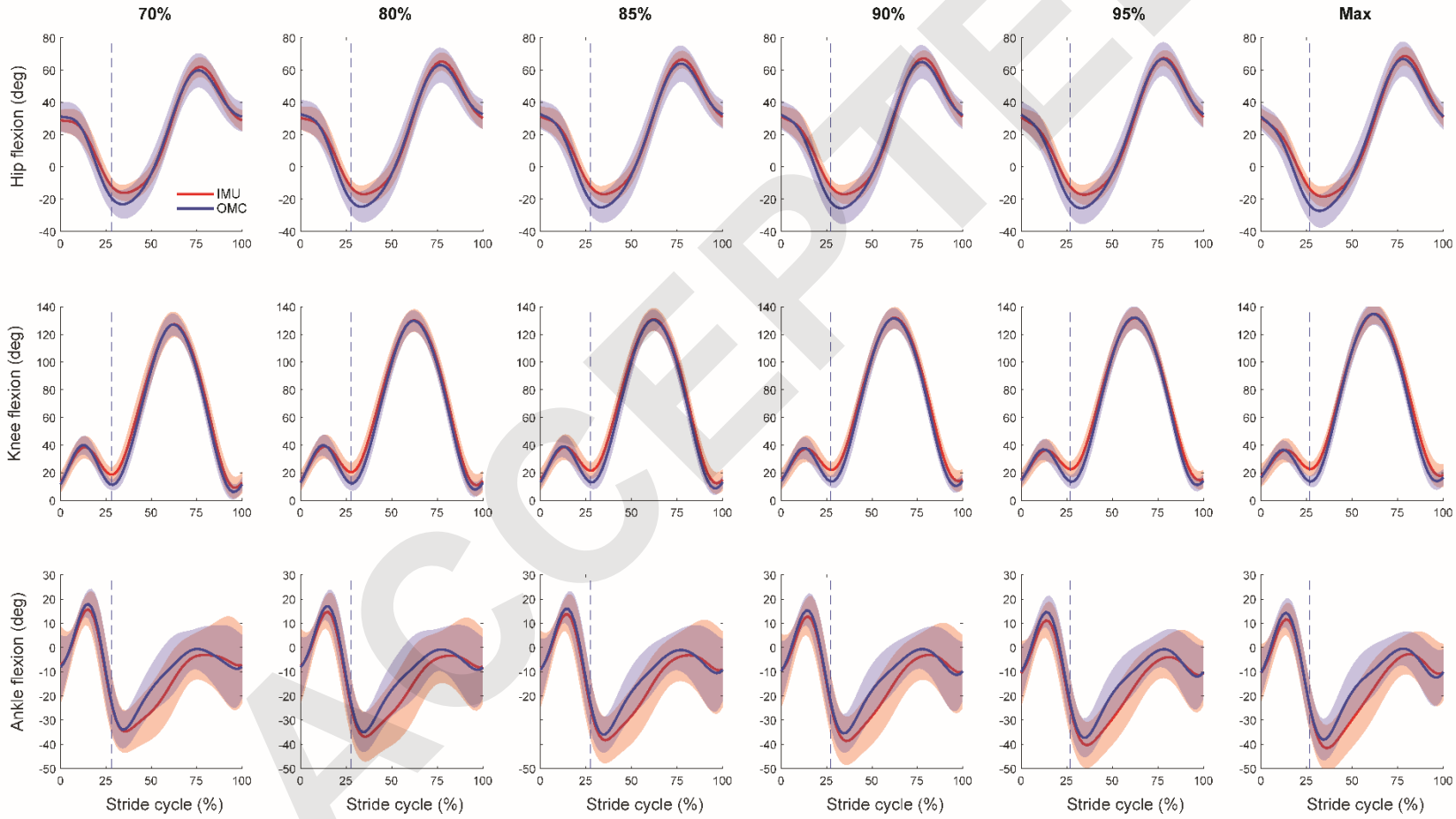
Figure 8



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

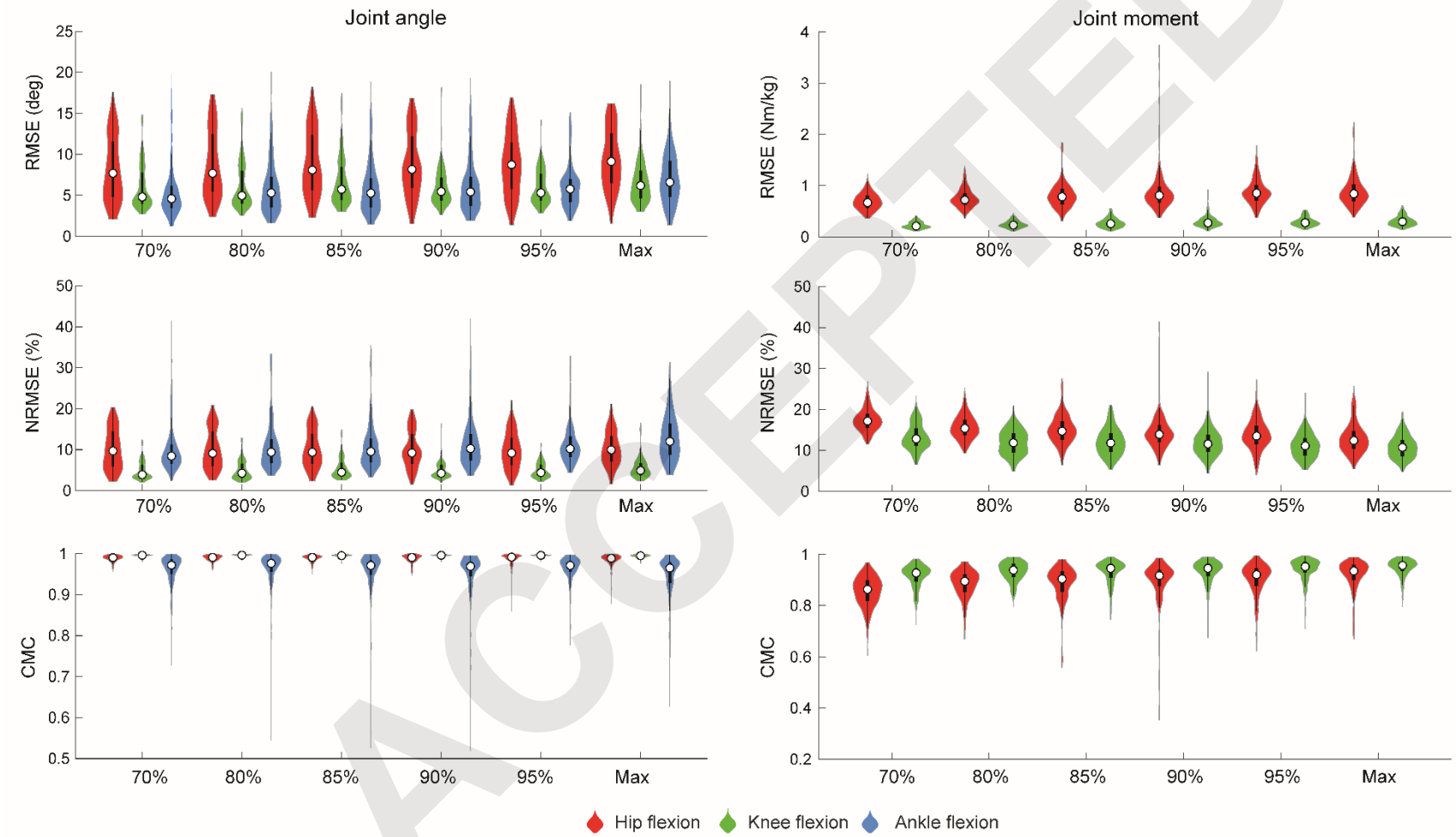
Supplemental Figure 1: Comparison of the sagittal-plane hip, knee, and ankle joint angles across one stride cycle, between IMU-based (red) and OMC-based (blue) measurements, at different running speeds (70%, 80%, 85%, 90%, 95%, and 100% of maximal running speed). All flexion angles are positive. The dashed vertical line indicates the end of the stance phase for each running speed. The shaded areas (red and blue) represent ± 1 standard deviation from the mean values.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Supplemental Figure 2: Violin plots illustrating the distributions of root mean square error (RMSE), normalised RMSE (NRMSE), and correlation coefficient (*r*) for sagittal-plane joint angles and moments between IMU- and OMC-based measurements at six different running speeds. The white dot represents the median, the black bar denotes the interquartile range, and the thin black line indicates the full data range.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

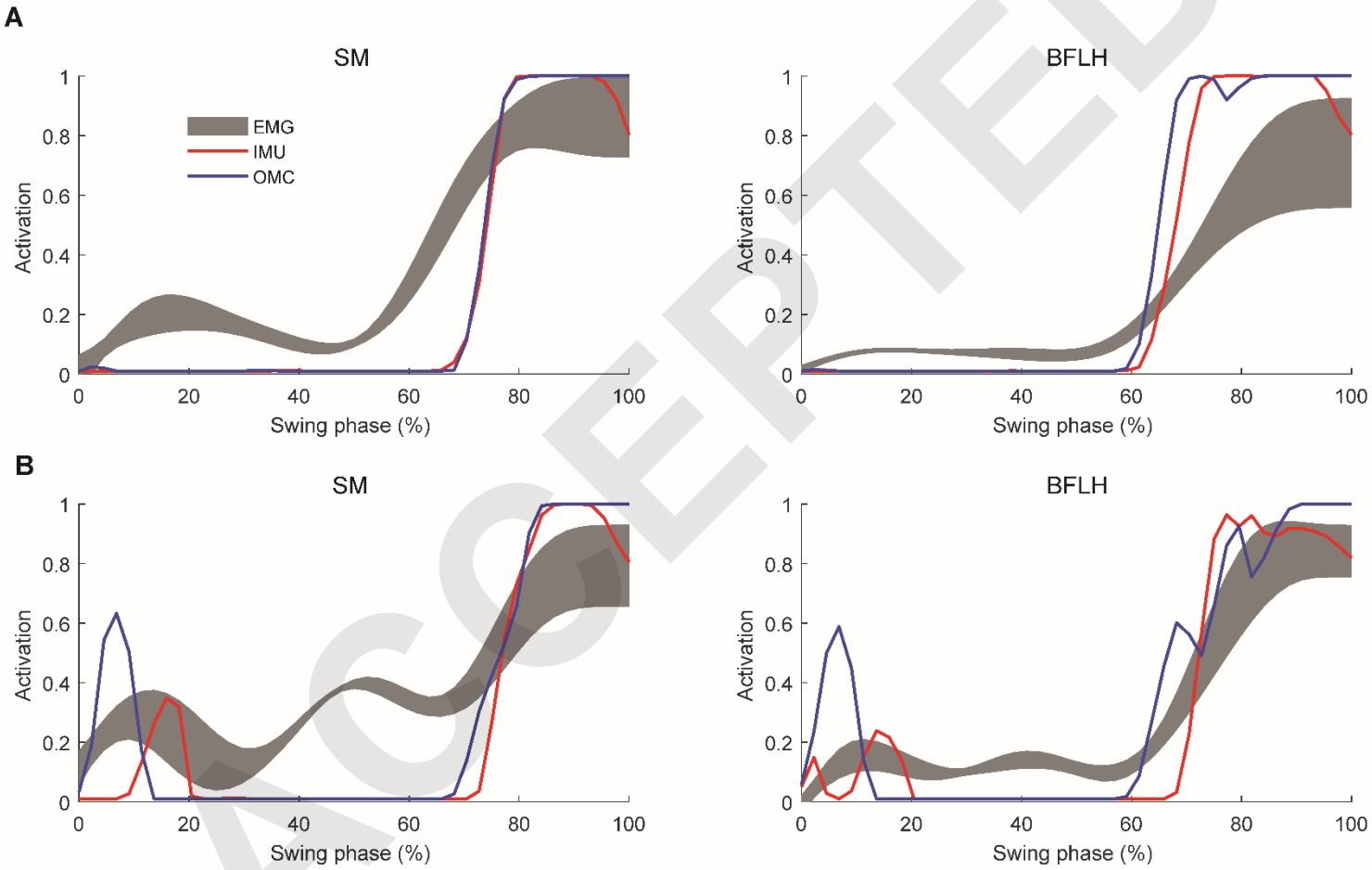
Supplemental Table 1: Linear mixed model results for the root mean square error (RMSE) of the sagittal-plane joint angles and moments, hamstring musculotendon forces, strains, and velocities with the peak running speed within a trial as the fixed effect and participant identification as the random effect.

				95% Confidence Interval	
Model	**Term**	**Estimate**	**p-value**	**Lower**	**Upper**
RMSE Hip angle (°)	Intercept (°)	5.59	<0.001	3.38	7.79
	Slope (°/(m·s^{-1}))	0.52	<0.001	0.31	0.74
RMSE Knee angle (°)	Intercept (°)	3.97	<0.001	2.36	5.57
	Slope (°/(m·s^{-1}))	0.38	<0.001	0.22	0.54
RMSE Ankle angle (°)	Intercept (°)	2.27	0.02	0.29	4.25
	Slope (°/(m·s^{-1}))	0.64	<0.001	0.38	0.9
RMSE Hip moment (Nm/kg)	Intercept (Nm/kg)	-0.35	<0.001	-0.46	-0.24
	Slope ((Nm/kg)/(m·s^{-1}))	0.23	<0.001	0.21	0.24
RMSE Knee moment (Nm/kg)	Intercept (Nm/kg)	-0.02	0.46	-0.08	0.04
	Slope ((Nm/kg)/(m·s^{-1}))	0.04	<0.001	0.04	0.05
RMSE BFLH MT force (BW)	Intercept (BW)	0.39	<0.001	0.36	0.41
	Slope (BW/(m·s^{-1}))	0.01	<0.01	0.002	0.01
RMSE BFSH MT force (BW)	Intercept (BW)	-0.05	<0.001	-0.06	-0.03
	Slope (BW/(m·s^{-1}))	0.03	<0.001	0.03	0.03
RMSE SEMIM MT force (BW)	Intercept (BW)	0.15	<0.001	0.11	0.19
	Slope (BW/(m·s^{-1}))	0.04	<0.001	0.03	0.04
RMSE SEMIT MT force (BW)	Intercept (BW)	-0.11	<0.001	-0.13	-0.09
	Slope (BW/(m·s^{-1}))	0.04	<0.001	0.03	0.04
RMSE BFLH MT strain (%)	Intercept (%)	1.54	<0.001	1.33	1.75
	Slope (%/(m·s^{-1}))	0.07	<0.001	0.05	0.1
RMSE BFSH MT strain (%)	Intercept (%)	1.73	<0.001	1.62	1.84
	Slope (%/(m·s^{-1}))	0.02	<0.001	0.01	0.03
RMSE SEMIM MT strain (%)	Intercept (%)	1.44	<0.001	1.27	1.6
	Slope (%/(m·s^{-1}))	0.06	<0.001	0.04	0.08
RMSE SEMIT MT strain (%)	Intercept (%)	1.57	<0.001	1.38	1.77
	Slope (%/(m·s^{-1}))	0.07	<0.001	0.05	0.09
RMSE BFLH MT velocity (m/s)	Intercept (m/s)	0.03	<0.001	0.02	0.04
	Slope ((m/s)/(m·s^{-1}))	0.01	<0.001	0.01	0.01
RMSE BFSH MT velocity (m/s)	Intercept (m/s)	-0.001	0.84	-0.01	0.004
	Slope ((m/s)/(m·s^{-1}))	0.01	<0.001	0.01	0.01
RMSE SEMIM MT velocity (m/s)	Intercept (m/s)	0.01	0.08	-0.001	0.02
	Slope ((m/s)/(m·s^{-1}))	0.01	<0.001	0.01	0.02
RMSE SEMIT MT velocity (m/s)	Intercept (m/s)	0.01	0.09	-0.002	0.03
	Slope ((m/s)/(m·s^{-1}))	0.02	<0.001	0.02	0.02

ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

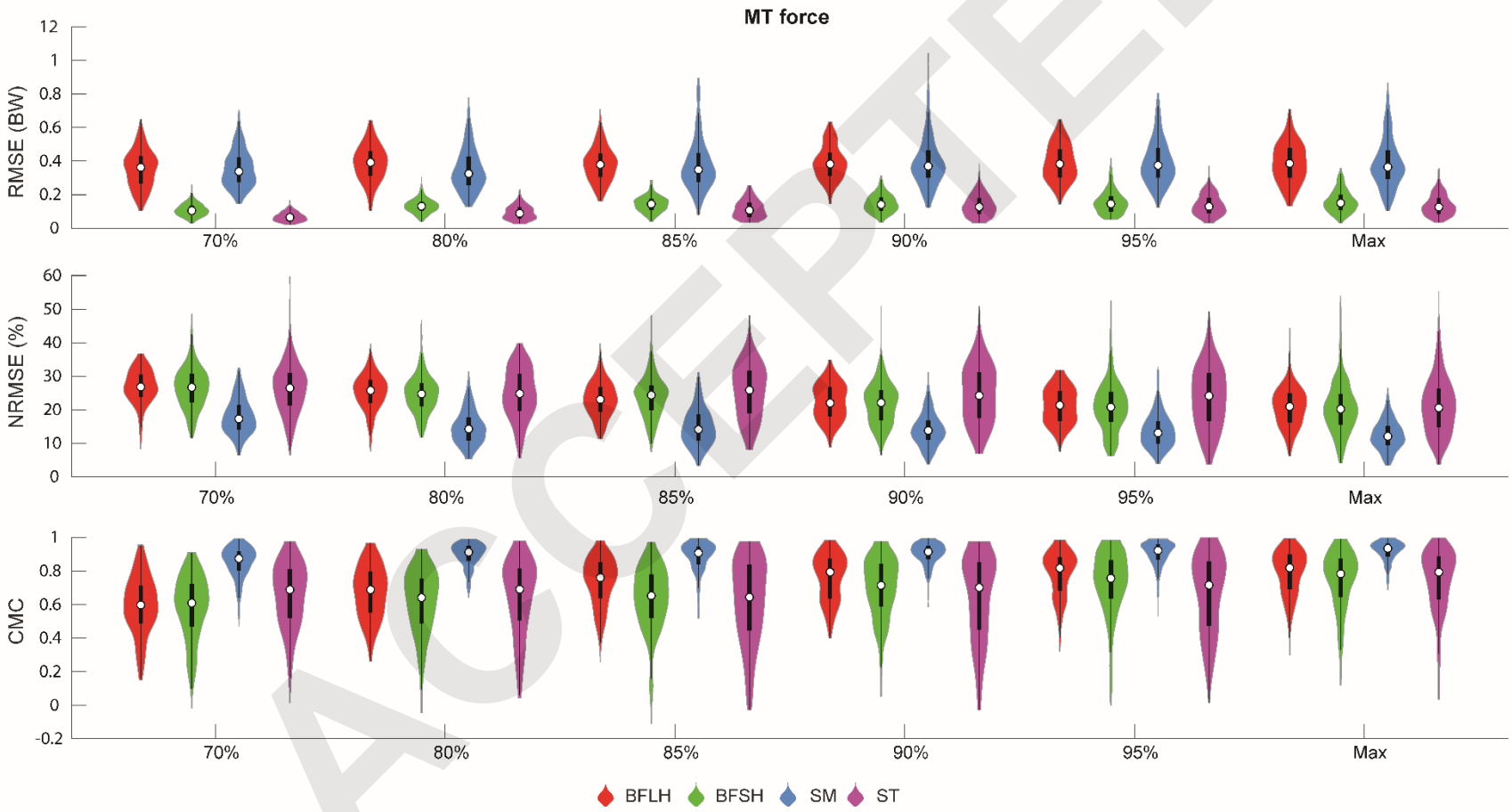
Supplemental Figure 3: Comparison of averaged EMG data and muscle activation data for SM and BFLH muscles in a representative male (A) and female (B) participant at the maximal running speeds. The shaded regions represent ± 1 standard deviation of the mean EMG. Muscle symbols are as follows: BFLH, biceps femoris long head; SM, semimembranosus.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

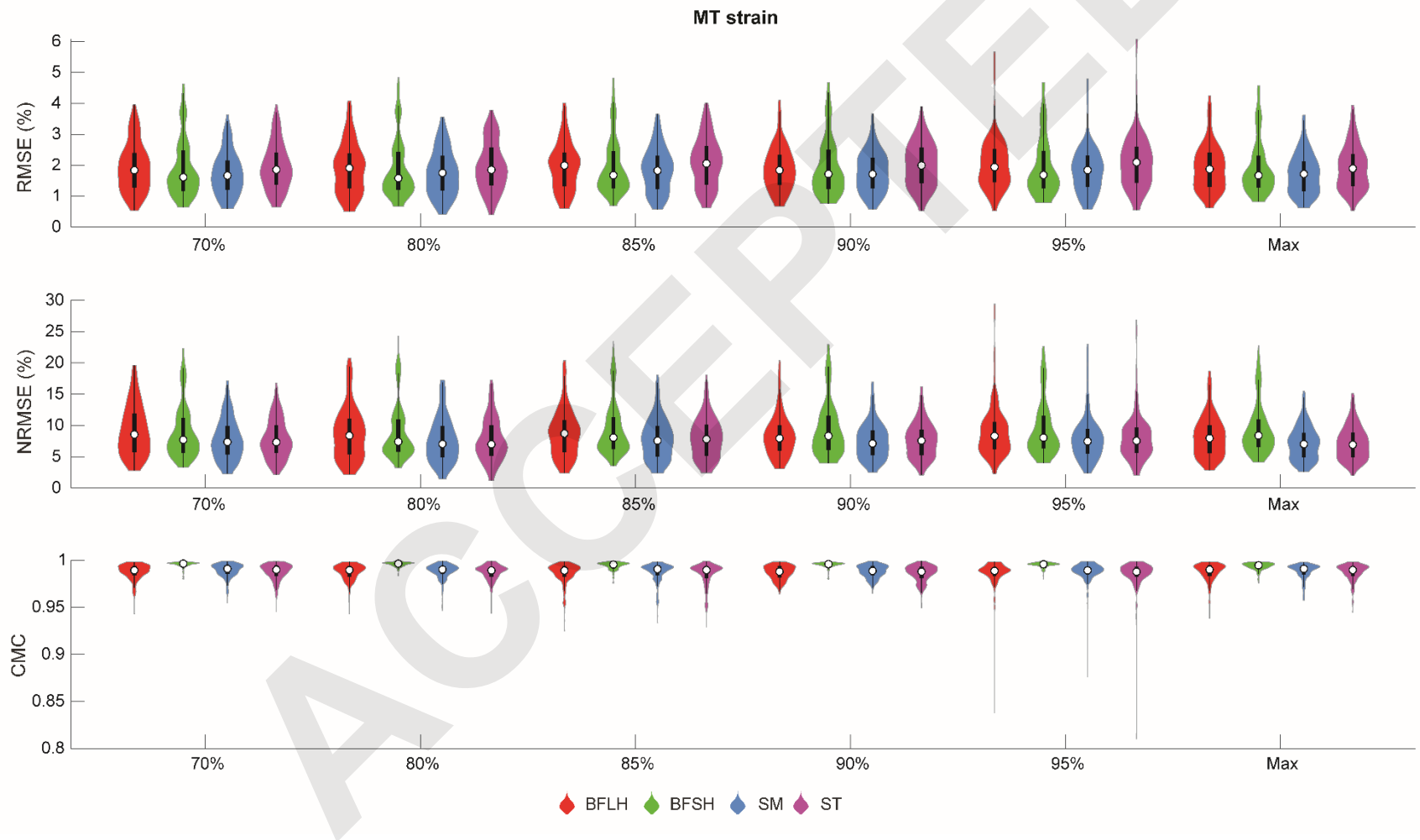
Supplemental Figure 4: Violin plots illustrating the distributions of root mean square error (RMSE), normalised RMSE (NRMSE), and coefficient of multiple correlation (CMC) for hamstring musculotendon forces between IMU- and OMC-based measurements at six different running speeds. The white dot represents the median, the black bar denotes the interquartile range, and the thin black line indicates the full data range. Muscle symbols are as follows: BFLH, biceps femoris long head; BFSH, biceps femoris short head; SM, semimembranosus; ST, semitendinosus.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

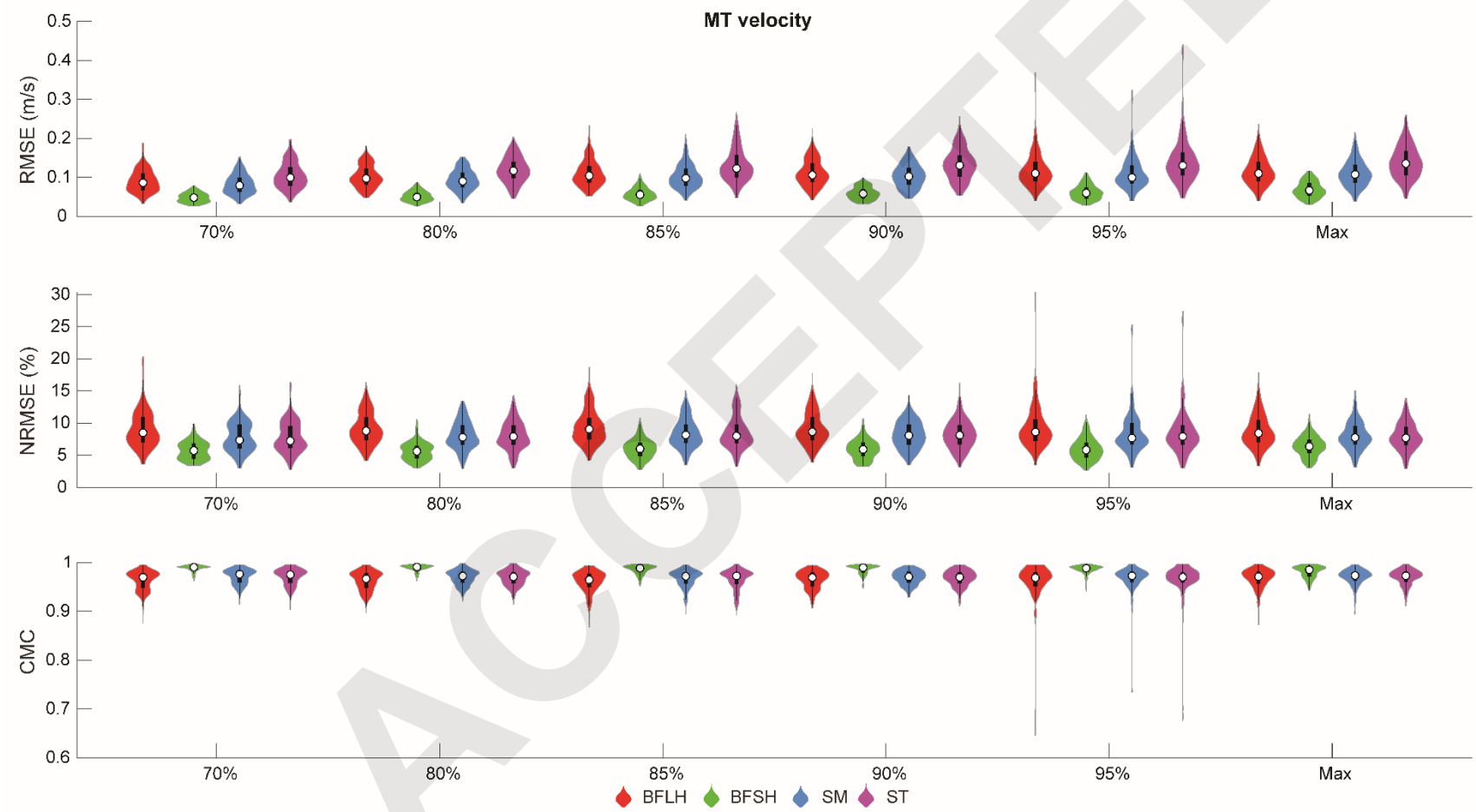
Supplemental Figure 5: Violin plots illustrating the distributions of root mean square error (RMSE), normalised RMSE (NRMSE), and coefficient of multiple correlation (CMC) for hamstring musculotendon strains between IMU- and OMC-based measurements at six different running speeds. The white dot represents the median, the black bar denotes the interquartile range, and the thin black line indicates the full data range. Muscle symbols are as follows: BFLH, biceps femoris long head; BFSH, biceps femoris short head; SM, semimembranosus; ST, semitendinosus.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

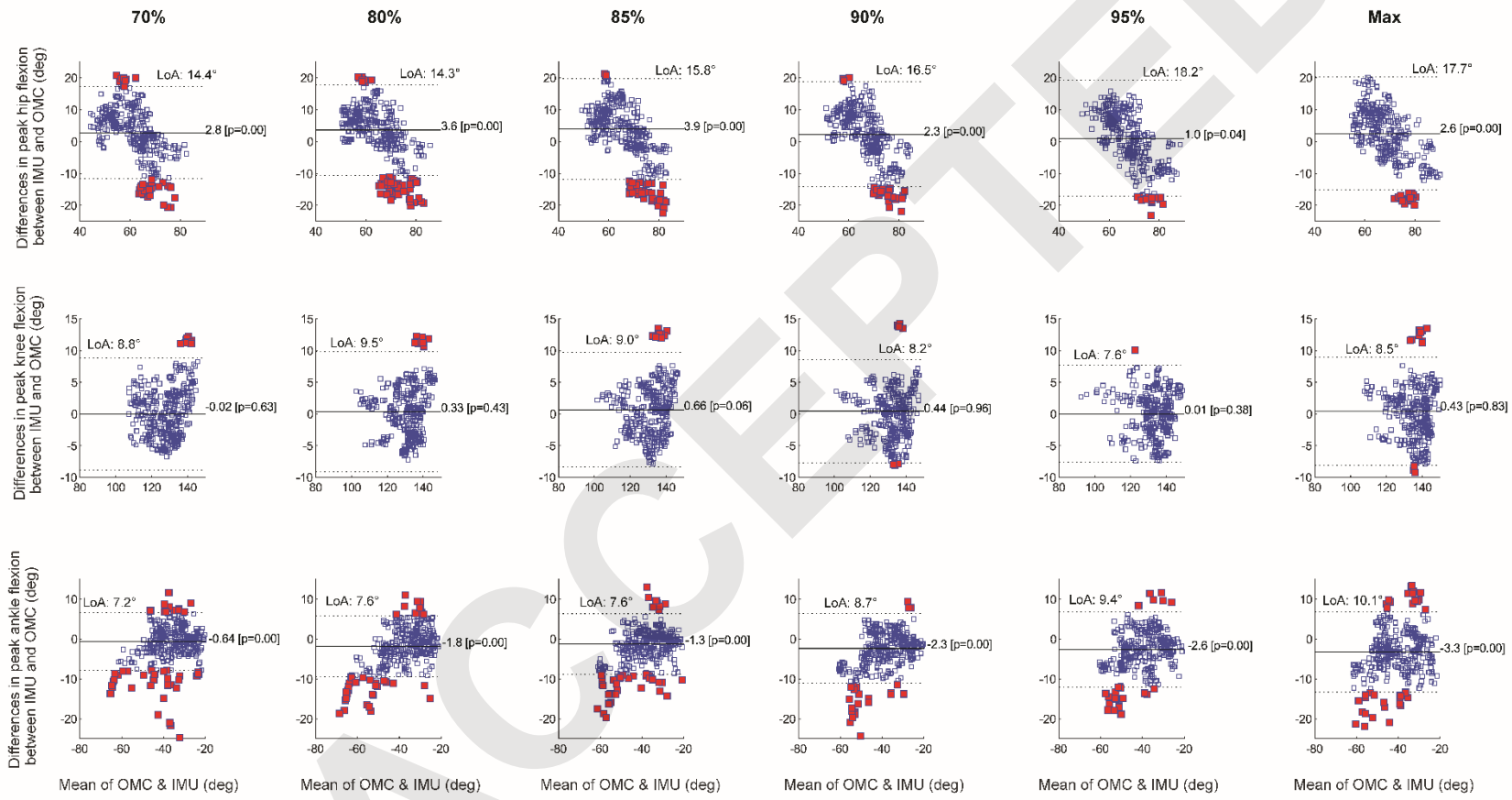
Supplemental Figure 6: Violin plots illustrating the distributions of root mean square error (RMSE), normalised RMSE (NRMSE), and coefficient of multiple correlation (CMC) for hamstring musculotendon velocities between IMU- and OMC-based measurements at six different running speeds. The white dot represents the median, the black bar denotes the interquartile range, and the thin black line indicates the full data range. Muscle symbols are as follows: BFLH, biceps femoris long head; BFSH, biceps femoris short head; SM, semimembranosus; ST, semitendinosus.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Supplemental Figure 7: Bland-Altman plots illustrate the peak hip flexion, knee flexion, and ankle flexion angles during running at six different speeds. Each square represents the average peak joint angle on the horizontal axis and the difference between the IMU and OMC systems on the vertical axis. The systematic bias (black horizontal line) and the 95% limits of agreement (black dotted lines) are shown in each plot. Data points outside the limits of agreement are represented by red squares. LoA = Limits of Agreement.



ACCEPTED

Copyright © 2025 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.